

Card column reference

All **711** columns across the five cards, grouped by card and then by block, with blocks ordered by **measured importance**. Each entry leads with what the column *is*; the caveats fold underneath.

Generated 2026-08-20 by `analyses/ops/gen_column_reference.py` from `card_glossary.tsv` and `card_registry.tsv`. **Do not hand-edit**.

Vintage mix “ read before quoting a fitted column

The delivered edge card is **16 days newer** than the base it layers over. Its **84 base-owned columns** (`coupling_`, `beta_`, `dose_`, `share_`, `rank_`) carry the older vintage; the other 102 are current.

The one rule behind this layout

The rung is a property of (CARD, COLUMN) “ never of the column name. `beta` is edge-rung on the edge card and family-rung on the `gene_family` card: same estimator, different unit. That is why this is organised card-first, and why one name can appear twice with different text.

How blocks are ordered

By three measured signals, not by size: how often a block's columns are named in the **discovery registry** (what the findings rest on, weighted highest), how many **modules** read them, and whether the block is **base-owned** (fitted) rather than an annotation join. Names under 6 characters count only backtick-quoted registry hits “ otherwise `n` and `z` match ordinary prose.

One correction the measurement needed: mention-frequency is not importance “ a *settled* block gets discussed less precisely because nobody argues about it. `adm_` scored 27th of 44 on the raw signals while MH-216 calls `adm_has_site` the single most load-bearing conditioning variable, on 4 registry mentions. Five blocks therefore carry a **curated floor**, each citing the finding that justifies it (see `CURATED_FLOOR`). That is the only hand-set input here.

Cards

- edge “ 186 columns, 44 blocks
- gene “ 135 columns, 29 blocks
- gene_family “ 61 columns, 12 blocks
- arm “ 296 columns, 43 blocks
- seed_family “ 33 columns, 2 blocks

edge

One row per one (miRNA arm, gene) pair. The widest card and the one most claims are made from. 84 of its columns come from the BASE card and 102 from the annotation layer “ check the vintage banner before quoting a fitted one.

186 columns 5,649 rows 44 blocks median coverage 72%

Contents “ 44 blocks, most important first

1. (bare) 12 “ CORE

2. coupling_ 13 “ CORE

3. arm_ 14 “ CORE

4. seed_ 1 “ CORE

5. adm_ 4 “ CORE

6. beta_ 8 “ CORE

7. n_ 4 “ CORE

8. cal_ 7 “ CORE

9. echim_ 3 “ CORE

10. share_ 4 “ CORE

11. rank_ 4 “ CORE

12. shift_ 1 “ CORE

13. ctx_ 18 “ SUPPORTING

14. family_ 4 “ SUPPORTING

15. healthy_ 3 “ SUPPORTING

16. identity_ 5 “ SUPPORTING

17. dose_ 7 “ SUPPORTING

18. d_ 4 “ SUPPORTING

19. ago_ 2 “ REFERENCE

20. gene_ 5 “ REFERENCE

21. retention_ 3 “ REFERENCE

22. pip_ 2 “ REFERENCE
23. oof_ 3 “ REFERENCE

24. grank_ 3 “ REFERENCE

25. own_ 1 “ REFERENCE

26. w_ 1 “ REFERENCE

27. composition_ 1 “ REFERENCE

28. m_ 1 “ REFERENCE

29. prior_ 1 “ REFERENCE

30. term_ 3 “ REFERENCE

31. edge_ 1 “ REFERENCE

32. mean_ 2 “ REFERENCE

33. surrogate_ 2 “ REFERENCE

34. esub_ 8 “ REFERENCE

35. wiring_ 1 “ REFERENCE

36. kd_ 2 “ REFERENCE

37. realization_ 2 “ REFERENCE

38. acquisition_ 1 “ REFERENCE

39. sd_ 1 “ REFERENCE

40. z_ 1 “ REFERENCE

41. heur_ 1 “ REFERENCE

42. cptac_ 20 “ REFERENCE

43. hly_ 1 “ REFERENCE

44. is_ 1 “ REFERENCE

(BARE) “ 12

core “ the columns findings rest on

arm KEY100%	KEY “ mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through <code>_arm_key()</code> , which gates against the canonical universe and LOGS drops rather than silently dropping.
beta EDGE100%	EDGE-rung coupling coefficient “ <code>readouts.run(level='arm')</code> , the same-seed collapse REMOVED.
dGlobal_HLY_NAT ARM63%	Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step. Defined on: arms with a GTeX-HEALTHY leg (the thinnest state “ many arms are multi-mapping-collapsed, MH-166)

dGlobal_HLY_TUM ARM63%	Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)
dGlobal_NAT_TUM ARM68%	Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step. Defined on: edges with a matched NAT leg (n~104 paired participants)
dShare_M_own EDGE65%	Mean over the 103 paired patients of this arm's change in M-WEIGHTED share of its gene's total regulator dose, tumour minus that patient's OWN NAT.
dShare_raw_own EDGE65%	As dShare_M_own but on RAW abundance share, with the model's M weights removed.
gene KEY100%	KEY â€” HGNC gene symbol.
identified EDGE100%	 z > 2 on the UNCALIBRATED posterior SD, where z = beta / beta_sd (MH-83 records precision 0.86 / recall 0.89 against a held-out standard). True on 24.8% of edges.
identity EDGE96%	Shapley/LMG share of the gene's R^2 credited to this edge, with bagged-NNLS weights.
role GENE	â€” NOT the arm's role â€” the GENE's cancer role (oncogene / tsg / oncogene/tsg / unknown), from gene_roles.load_gene_roles .
z EDGE100%	beta / beta_sd for the EDGE-level fit (readouts.run(level='arm') , the whole gene's arms as separate predictors), on the UNCALIBRATED posterior SD.
COUPLING_ Â· 13 core â€” the columns findings rest on	
coupling_fam FAMILY100%	Family-level coupling for the edge's (gene, seed_family) cell.
coupling_hly EDGE56%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_hly_surrogate EDGE3%	Healthy coupling via a same-seed SURROGATE arm, used where the arm itself has no GTEx signal.
coupling_nat EDGE62%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_p_hly EDGE56%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_p_hly_resolved EDGE59%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_p_hly_surrogate EDGE3%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_p_nat EDGE62%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.

coupling_p_tum EDGE 69%	p against the CALIBRATED site-free null (not a theoretical t-null). Median 0.375; <0.05 on 16.2%.
coupling_tum EDGE 72%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_z_hly EDGE 59%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_z_nat EDGE 62%	Partial-Spearman coupling of arm dose against target mRNA in one state, conditioned on C (composition + target CN + proliferation), with its p and z.
coupling_z_tum EDGE 69%	The tumour coupling expressed in NULL SDs â€” the CALIBRATED effect size, and the scale to quote. Median $\hat{\alpha}^*0.32$; z >2 on 15.5%, z >3 on 6.4%.

ARM_ 14

core â€” the columns findings rest on

arm_credit_share ARM-IN-FAMILY 20%	This arm's share of its (gene, seed_family) cell's credit. â€” Verified to sum to 1.0000 within every one of the 467 multi-arm cells, and it genuinely varies PER ARM inside the cell (467/467). Defined on: multi-arm cells only ($n_{arm_in_cell} > 1$) â€” 20.3% of edges
arm_dbeta FAMILY 20%	$\max \hat{\alpha}^* \min \hat{\alpha}^2$ across the cell's member arms.
arm_detection ARM 95%	Fraction of patients in which this ARM is measured at all (<code>readouts._detection</code>).
arm_id_status ARM-IN-FAMILY 100%	How the arm's identity within its cell was resolved: <code>resolved_by_dose</code> 2,344 $\hat{\alpha}^*$ <code>singleton</code> 2,260 $\hat{\alpha}^*$ <code>unvalidated_balanced</code> 1,045.
arm_iqr ARM 69%	Interquartile range of the arm's abundance â€” its DYNAMIC RANGE.
arm_lfc_HLY_NAT_raw ARM 63%	log-FC GTEx-healthy $\hat{\alpha}^*$ TCGA-NAT, raw.
arm_lfc_HLY_TUM_QN ARM 64%	log-FC GTEx-healthy $\hat{\alpha}^*$ TCGA-tumour, on quantile-normalised values.
arm_lfc_HLY_TUM_raw ARM 63%	As <code>arm_lfc_HLY_TUM_QN</code> but against the RAW GTEx median.
arm_lfc_NAT_TUM ARM 68%	log-FC of the arm's median abundance, matched NAT $\hat{\alpha}^*$ tumour.
arm_med_rpm ARM 69%	Median linear RPM of this ARM across the cohort. Arm-rung, so it REPEATS across the arm's genes. Fill 0.686 â€” the gap is arms with no expression record, not zeros. Defined on: edges present in the progression panel (arm expressed + state measured)
arm_pct_floor ARM	Percent of samples in which the arm sits above the detection floor (median 99). Feeds <code>arm_spiker</code> together with <code>arm_iqr</code> . Defined on: edges present in the progression panel (arm expressed + state measured)
arm_resolvable FAMILY	Are the cell's arms separable AND does splitting them help out-of-fold. True for 31.7% of edges in multi-arm cells.

arm_sep_z	arm_dbeta in units of the cell's own pooled posterior SD “ can these same-seed arms be told apart at all.
FAMILY 20%	

arm_spiker	arm_pct_floor < 40 AND arm_iqr > 1.5 “ the arm sits at/below the detection floor in most samples yet swings widely when present.
ARM 69%	

SEED_ 1

core “ the columns findings rest on

seed_family	KEY “ the seed family itself, one row per family.
FAMILY 100%	

ADM_ 4

core “ the columns findings rest on

adm_admissible	The admissibility verdict: site + evidence + expression.
EDGE 87%	

adm_expr_gtex	ADMISSIBILITY “ could this edge repress in breast AT ALL? has_site (seed match in the 3'UTR) AND expressed (arm present in TCGA tumour OR GTEx healthy breast).
EDGE 87%	

adm_expr_tcga	Is the arm expressed above floor in TCGA?
EDGE 87%	

adm_has_site	Does the arm have a seed match in this target's 3'UTR.
EDGE 87%	

BETA_ 8

core “ the columns findings rest on

beta_arm	The arm's OWN $\hat{\tau}^2$ from the within-cell fit (arm_rung.py), defined only in multi-arm cells (fill 0.203).
EDGE 20%	

beta_deconv	$\hat{\tau}^2$ refit on the DECONV design (core C + the 8 non-malignant Wu-major lineages, Cancer-Epithelial held out).
EDGE 100%	

beta_frac	MAGNITUDE share (beta_f / sum beta).
EDGE 100%	

beta_frac_abs	$ E[\hat{\tau}_f^2] / \hat{\tau} E[\hat{\tau}^2]$ “ the share computed from the POINT ESTIMATES.
EDGE 100%	

beta_frac_deconv	beta_frac recomputed on the deconv design “ the composition-adjusted share of the gene's budget.
EDGE 100%	

beta_frac_sd	Posterior SD of the per-draw fraction $\hat{\tau}_f / \hat{\tau}$.
EDGE 100%	

beta_sd	Posterior summaries of the coupling coefficient from the dense Gibbs fit (learned/readouts.py). _sd is the posterior SD, _frac the share of the gene's total beta.
EDGE 100%	

beta_sd_deconv	Posterior SD on the deconv design.
EDGE	100%

N_ 4

core the columns findings rest on

n_HLY_meas	How many of this gene's arms have a MEASURED healthy (GTEx) level as opposed to one manufactured by the multi-mapping collapse.
EDGE	72%
n_arm_in_cell	How many arms sit in this edge's (gene, seed_family) CELL i.e. how many of that family's members appear in THIS gene's design.
FAMILY	20%
n_fam	How many families compete at this edge's GENE. Gene-rung it repeats across the gene's edges by construction. beta is mathematically inert where it is 1.
GENE	100%
n_samples	Samples the edge was fit on CONSTANT (1,040) across the card.
GENE	100%

CAL_ 7

core the columns findings rest on

cal_beta_sd	CALIBRATED posterior width (<code>learned/calibration.py</code>): the reported SD corrected for the measured 1.181.34x narrowness, with <code>_lo / _hi</code> spanning the correction constant's own +/-1 SD.
EDGE	100%
cal_beta_sd_hi	CALIBRATED posterior width (<code>learned/calibration.py</code>): the reported SD corrected for the measured 1.181.34x narrowness, with <code>_lo / _hi</code> spanning the correction constant's own +/-1 SD.
EDGE	100%
cal_beta_sd_lo	CALIBRATED posterior width (<code>learned/calibration.py</code>): the reported SD corrected for the measured 1.181.34x narrowness, with <code>_lo / _hi</code> spanning the correction constant's own +/-1 SD.
EDGE	100%
cal_identified	Is the edge identified (<code> cal_z > 2</code>) under the CALIBRATED posterior width the honest version of <code>identified</code> : 1,110 vs 1,375 edges , i.e. 19.9% [18.122.4] against 24.8%.
EDGE	100%
cal_identified_hi	CALIBRATED posterior width (<code>learned/calibration.py</code>): the reported SD corrected for the measured 1.181.34x narrowness, with <code>_lo / _hi</code> spanning the correction constant's own +/-1 SD.
EDGE	100%
cal_identified_lo	CALIBRATED posterior width (<code>learned/calibration.py</code>): the reported SD corrected for the measured 1.181.34x narrowness, with <code>_lo / _hi</code> spanning the correction constant's own +/-1 SD.
EDGE	100%
cal_z	CALIBRATED posterior width (<code>learned/calibration.py</code>): the reported SD corrected for the measured 1.181.34x narrowness, with <code>_lo / _hi</code> spanning the correction constant's own +/-1 SD.
EDGE	100%

ECHIM_ 3

core the columns findings rest on

echim_manakov_w	CHIMERIC ligation evidence at its NATIVE (gene, arm) rung a physical duplex read, per source, never pooled.
EDGE	22%

echim_n_sources EDGE26%	How many chimeric-eCLIP sources support this edge (1 on 1,151 · 2 on 262 · 3 on 34).
echim_tarbase_w EDGE8%	CHIMERIC ligation evidence at its NATIVE (gene, arm) rung · a physical duplex read, per source, never pooled.
SHARE_ · 4 core · the columns findings rest on	
share_HLY EDGE72%	The arm's share of the gene's healthy (GTEx) pressure budget, median 0.129 .
share_HLY_meas EDGE46%	share_HLY over ONLY the arms GTEx genuinely measures · invisible arms ABSTAIN (NaN) instead of being zeroed (MH-210). Median 0.244 against the imputed 0.129.
share_NAT EDGE72%	The arm's share of the gene's total regulatory dose in one state (HLY / NAT / TUM). Within-gene, sums to 1 across the gene's arms. Defined on: edges with a matched NAT leg (n~104 paired participants)
share_TUM EDGE72%	The arm's share of the gene's total regulatory dose in one state (HLY / NAT / TUM). Within-gene, sums to 1 across the gene's arms. Defined on: edges present in the progression panel (arm expressed + state measured)
RANK_ · 4 core · the columns findings rest on	
rank_HLY EDGE72%	The arm's WITHIN-GENE rank by dose in one state.
rank_HLY_meas EDGE46%	The arm's WITHIN-GENE rank by dose in one state.
rank_NAT EDGE72%	The arm's WITHIN-GENE rank by dose in one state.
rank_TUM EDGE72%	The arm's WITHIN-GENE rank by dose in one state.
SHIFT_ · 1 core · the columns findings rest on	
shift_class EDGE72%	Cross-state class for the edge.
CTX_ · 18 supporting · read these when the core raises a question	
ctx_apriori_class GENE100%	Gene-design context from the decoy/atlas layer (learned/analyses/gene_atlas.py via card_context): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

ctx_arm_abundant ARM 95%	Is this ARM above the design's abundance threshold.
ctx_arm_dose ARM 95%	The ARM's dose in this gene's design context.
ctx_ceiling GENE 99%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_collapse GENE 100%	Mean arms collapsed per family in this gene's design (median 1.07) â€” how much the seed-family collapse actually merged. Near 1 means the collapse was almost a no-op for this gene.
ctx_d_collin GENE 84%	Within-design COLLINEARITY asymmetry, real minus decoy. The second design-match axis MH-167 tested; also bounded and not moving the gap. <i> Defined on: genes with >=2 seed families (a collinearity measure needs 2 columns)</i>
ctx_d_fam GENE 99%	Design-WIDTH asymmetry between the real design and its matched decoy (real minus fake families).
ctx_dose_max GENE 100%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_dose_med GENE 100%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_frac_abund GENE 100%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_gap_core GENE 99%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_gap_d_dose GENE 99%	Dose asymmetry between the real and decoy designs (median $\hat{\alpha}^{0.43}$).
ctx_gap_deconv GENE 99%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_gap_retention GENE 95%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_measurable GENE 99%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_n_abund GENE 100%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_n_fam_abund GENE 100%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

ctx_n_fam_multi GENE 100%	Gene-design context from the decoy/atlas layer (learned/analyses/gene_atlas.py via card_context): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
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FAMILY_ 4
supporting read these when the core raises a question

family_dose_share ARM-IN-FAMILY	The arm's share of its seed family's TOTAL dose denominator is the full family , not the design (median 0.69). The quantity family_role is cut from. <i>Defined on: edges present in the progression panel (arm expressed + state measured)</i>
family_rho_adj FAMILY 69%	Family-level properties carried onto this row (size, the arm's dose share of it, and its role).
family_role ARM-IN-FAMILY	The ARM's DOSE role inside its seed family NOT the family's role in the gene, which is what the name suggests.
family_size FAMILY	Members of this arm's seed family THAT ARE IN THIS GENE'S DESIGN 1 on 4,497 of 5,644 edges (79.7%) .

HEALTHY_ 3
supporting read these when the core raises a question

healthy_leg ARM 72%	Healthy-state legs and flags including healthy_uninformative , which marks rows where the healthy baseline cannot carry a claim. <i>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</i>
healthy_potential ARM 64%	Healthy-state legs and flags including healthy_uninformative , which marks rows where the healthy baseline cannot carry a claim. <i>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</i>
healthy_uninformative ARM 64%	Healthy-state legs and flags including healthy_uninformative , which marks rows where the healthy baseline cannot carry a claim. <i>Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)</i>

IDENTITY_ 5
supporting read these when the core raises a question

identity_abs EDGE 96%	identity as a fraction of the gene's total identity , clipped to [0,1] .
identity_allocated EDGE 96%	Family identity with phantom minor arms forced to 0 the allocation actually used downstream.
identity_coherence EDGE 96%	1 / Î£ identity over the gene, clipped to (0,1] how much the gene's identity shares CANCEL.
identity_deconv EDGE 98%	Shapley identity recomputed under the composition-adjusted design.

identity_reliable	Is this edge's Shapley <code>identity</code> trustworthy â€” <code>identity_coherence</code> â‰¥ 0.5 AND identity â‰¥ 1?
EDGE	96%

DOSE_ 7
supporting â€” read these when the core raises a question

dose_class	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. Defined on: edges present in the progression panel (arm expressed + state measured)
EDGE	72%
dose_comp_retention	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. Defined on: edges with a matched NAT leg (n~104 paired participants)
ARM	55%
dose_confounded	Is this edge's NATâ€™tumour dose a composition/proliferation artifact (either gated retention < 0.4)?
ARM	72%
dose_prolif_retention	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. Defined on: edges with a matched NAT leg (n~104 paired participants)
ARM	55%
dose_rank_HLY	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)
EDGE	72%
dose_rank_NAT	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. Defined on: edges with a matched NAT leg (n~104 paired participants)
EDGE	72%
dose_rank_TUM	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. Defined on: edges present in the progression panel (arm expressed + state measured)
EDGE	72%

D_ 4
supporting â€” read these when the core raises a question

d_rank_HLY_NAT	Change in the arm's WITHIN-GENE dose rank between two states.
EDGE	72%
d_rank_HLY_TUM	Change in the arm's WITHIN-GENE dose rank between two states.
EDGE	72%
d_rank_HLY_TUM_meas	<code>d_rank_HLY_TUM</code> recomputed over arms with a MEASURED healthy level only â€” the collapse-safe version. Prefer it over the bare column. Defined on: edges whose gene has â‰¥1 GTEx-MEASURED regulator â€” âŒ™ NaN = GTEx cannot see it (abstention), NOT 0
EDGE	46%
d_rank_NAT_TUM	Change in the arm's WITHIN-GENE dose rank between two states.
EDGE	72%

AGO_ 2

reference “ context, provenance and rarely-quoted detail

ago_loading ARM95%	AGO/RISC loading discount.
ago_loading_measured ARM100%	Was ago_loading actually MEASURED for this edge (vs defaulted to 1.0). True for 3,820/5,648 edges (67.6%). Defined on: arms with a Manakov AGO-dominance measurement “ NOT loading(), which fabricates 1.0

GENE_ 5

reference “ context, provenance and rarely-quoted detail

gene_dominated GENE97%	Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.
gene_iqr GENE72%	Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction. Defined on: genes with a NAT-TUM gene-level leg
gene_lfc_NAT_TUM GENE72%	Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction. Defined on: genes with a NAT-TUM gene-level leg
gene_net_repr GENE100%	Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.
gene_repr_class GENE100%	Gene-level properties carried onto this row (its regulator count, repression class, net-repression flag). On the edge card these REPEAT within gene by construction.

RETENTION_ 3

reference “ context, provenance and rarely-quoted detail

retention_beta EDGE100%	Fraction of an effect surviving covariate adjustment (adjusted / raw).
retention_reliable EDGE100%	Is this edge's retention ratio trustworthy “ i.e. does it have a real denominator?
retention_rho EDGE72%	$\tilde{I}_{deconv} / \tilde{I}_{core}$ “

PIP_ 2

reference “ context, provenance and rarely-quoted detail

pip_dense EDGE100%	“ CONSTANT by construction “ the coupling readout is called with $\tilde{I}_{\pi=1}$, so every edge has the same value.
pip_discovery EDGE100%	Posterior inclusion probability. pip_dense comes from the $\pi=1$ coupling readout; pip_discovery from the evidence- π readout.

OOF_ 3

reference “ context, provenance and rarely-quoted detail

oof_rho FAMILY 20%	Arm-level OOF rho MINUS family-level.
oof_rho_arm FAMILY 20%	Out-of-fold predictive correlation, arm-level vs family-level, and their difference (oof_rho = arm minus family). Negative = the arm-resolved fit predicts repression better.
oof_rho_fam FAMILY 20%	Out-of-fold predictive correlation, arm-level vs family-level, and their difference (oof_rho = arm minus family). Negative = the arm-resolved fit predicts repression better.

GRANK_ 3

reference “ context, provenance and rarely-quoted detail

grank_HLY ARM 63%	The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state “ many arms are multi-mapping-collapsed, MH-166)
grank_NAT ARM 68%	The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from. Defined on: edges with a matched NAT leg (n~104 paired participants)
grank_TUM ARM 69%	The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from. Defined on: edges present in the progression panel (arm expressed + state measured)

OWN_ 1

reference “ context, provenance and rarely-quoted detail

own_specific_frac ARM 88%	Fraction of the arm's variance that is target-specific.
--	---

W_ 1

reference “ context, provenance and rarely-quoted detail

w_max GENE 100%	Maximum curated EVIDENCE weight over the gene's regulators. Gene-rung repeat.
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COMPOSITION_ 1

reference “ context, provenance and rarely-quoted detail

composition_class EDGE 100%	Composition-adjustment verdict for this row (e.g. composition_explained when the effect does not survive the deconvolution block).
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M_ Â· 1

reference “ context, provenance and rarely-quoted detail

m_nnl

EDGE 100%

Bagged-NNLS weight. Exactly 0 in 31.7% of families “ this is what lets identity say a regulator contributes nothing, which beta structurally cannot.

PRIOR_ Â· 1

reference “ context, provenance and rarely-quoted detail

prior_pi

EDGE 100%

The Gibbs sampler's inclusion prior “ CONSTANT 0.05 across the card.

TERM_ Â· 3

reference “ context, provenance and rarely-quoted detail

term_ABUND

EDGE 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms.
| Defined on: edges present in the progression panel (arm expressed + state measured)

term_INTERACT

EDGE 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms.
| Defined on: edges present in the progression panel (arm expressed + state measured)

term_WIRING

EDGE 68%

Decomposition of the acquisition score into ABUNDANCE, WIRING and INTERACTION terms.
| Defined on: edges present in the progression panel (arm expressed + state measured)

EDGE_ Â· 1

reference “ context, provenance and rarely-quoted detail

edge_rho_adj

EDGE 73%

An edge-rung quantity carried onto this row.
| Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

MEAN_ Â· 2

reference “ context, provenance and rarely-quoted detail

mean_dGlobalRank

ARM 90%

A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises.
| Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

mean_own_shift

ARM 90%

A per-row mean of a lower-rung quantity; check agg_of for which rung it summarises.
| Defined on: the WITHIN-PATIENT paired subset (n=103 matched tumour/NAT pairs, MH-158)

SURROGATE_ Â· 2

reference “ context, provenance and rarely-quoted detail

surrogate_corr

ARM 3%

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for.
| Defined on: arms with a same-seed SURROGATE (MH-166) “ 3.6% of edges

surrogate_instrument

ARM 3%

Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for.
| Defined on: arms with a same-seed SURROGATE (MH-166) “ 3.6% of edges

ESUB_ 8

reference context, provenance and rarely-quoted detail

esub_best_sub EDGE100%	PAM50 subtype-stratified coupling at the EDGE rung one fit per subtype plus the pooled fit and an adjusted q.
esub_q_adj EDGE98%	BH-adjusted q for the best-subtype contrast
esub_rho_Basal EDGE99%	PAM50 subtype-stratified coupling at the EDGE rung one fit per subtype plus the pooled fit and an adjusted q.
esub_rho_Her2 EDGE98%	PAM50 subtype-stratified coupling at the EDGE rung one fit per subtype plus the pooled fit and an adjusted q.
esub_rho_LumA EDGE99%	PAM50 subtype-stratified coupling at the EDGE rung one fit per subtype plus the pooled fit and an adjusted q.
esub_rho_LumB EDGE99%	PAM50 subtype-stratified coupling at the EDGE rung one fit per subtype plus the pooled fit and an adjusted q.
esub_rho_best EDGE100%	Coupling in the arm's BEST PAM50 subtype.
esub_rho_pool EDGE100%	Coupling pooled across subtypes the unselected reference for esub_rho_best (median 0.0135, indistinguishable from a random-subtype draw at 0.0123).

WIRING_ 1

reference context, provenance and rarely-quoted detail

wiring_frac EDGE68%	Fraction of the acquisition score attributable to WIRING rather than abundance. Defined on: edges present in the progression panel (arm expressed + state measured)
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KD_ 2

reference context, provenance and rarely-quoted detail

kd_affinity_pct EDGE85%	Is this gene among the arm's strongest predicted targets a WITHIN-ARM percentile, higher = stronger target .
kd_repression EDGE85%	scanMiR RBNS binding affinity.

REALIZATION_ 2

reference context, provenance and rarely-quoted detail

realization_identity EDGE58%	Within-patient realization quantities the score, its identity allocation, and who owns it. Defined on: genes with a family-level realization fit
realization_score EDGE69%	Within-patient realization quantities the score, its identity allocation, and who owns it. Defined on: edges present in the progression panel (arm expressed + state measured)

ACQUISITION_ 1

reference context, provenance and rarely-quoted detail

acquisition_score	Composite acquisition score across states.
EDGE59%	Defined on: arms with a GTEx-HEALTHY leg (the thinnest state many arms are multi-mapping-collapsed, MH-166)

SD_ 1

reference context, provenance and rarely-quoted detail

sd_arm	Posterior SD of the arm-level $\hat{\tau}^2$.
EDGE20%	

Z_ 1

reference context, provenance and rarely-quoted detail

z_arm	<code>beta_arm / sd_arm</code> from the WITHIN-CELL fit (<code>arm_rung.py</code>) the arm's z inside its (gene, seed_family) cell only, defined on the 20.3% of edges in multi-arm cells.
EDGE20%	

HEUR_ 1

reference context, provenance and rarely-quoted detail

heur_gene_nreg	Regulator count for this gene from the $\S6b$ -RETIRED heuristic lane.
GENE100%	

CPTAC_ 20

reference context, provenance and rarely-quoted detail

cptac_prosp_comp_class_prot	Composition class of this edge's PROTEIN coupling (<code>cell_intrinsic / partial / composition_explained</code>) in the prospective cohort.
EDGE59%	
cptac_prosp_n	Patients in the prospective CPTAC cohort CONSTANT 101.
EDGE79%	
cptac_prosp_ret_prot	CPTAC protein channel.
EDGE59%	
cptac_prosp_ret_rna	CPTAC protein channel.
EDGE61%	
cptac_prosp_rho_disc	CPTAC protein channel.
EDGE78%	
cptac_prosp_rho_disc_raw	CPTAC protein channel.
EDGE78%	
cptac_prosp_rho_prot	CPTAC protein channel.
EDGE78%	
cptac_prosp_rho_prot_raw	UNADJUSTED protein coupling, prospective.
EDGE79%	

cptac_prosp_rho_rna	CPTAC protein channel.
EDGE 79%	
cptac_prosp_rho_rna_raw	CPTAC protein channel.
EDGE 79%	
cptac_t105_comp_class_prot	CPTAC protein channel.
EDGE 45%	
cptac_t105_n	Patients in the TCGA-overlap CPTAC cohort are CONSTANT 105 . See the prospective twin.
EDGE 70%	
cptac_t105_ret_prot	CPTAC protein channel.
EDGE 45%	
cptac_t105_ret_rna	CPTAC protein channel.
EDGE 49%	
cptac_t105_rho_disc	CPTAC protein channel.
EDGE 64%	
cptac_t105_rho_disc_raw	CPTAC protein channel.
EDGE 65%	
cptac_t105_rho_prot	CPTAC protein channel.
EDGE 64%	
cptac_t105_rho_prot_raw	CPTAC protein channel.
EDGE 65%	
cptac_t105_rho_rna	CPTAC protein channel.
EDGE 64%	
cptac_t105_rho_rna_raw	CPTAC protein channel.
EDGE 65%	

HLY_ 1

reference are context, provenance and rarely-quoted detail

hly_leg_concordant	Do the two HLY’s TUM legs (_qn and _raw) at least AGREE ON SIGN. Concordant 76.1%, DISCORDANT 23.9% of the 3,433 edges carrying both.
EDGE 61%	

IS_ 1

reference are context, provenance and rarely-quoted detail

is_realization_owner	A boolean flag; read the suffix. <i>Defined on: genes with a family-level realization fit</i>
EDGE 58%	

gene

One row per the gene's total incoming regulation. One row per gene, so nothing here is checkable by invariance – `agg_of` carries the meaning instead.

135 columns 1,549 rows 29 blocks median coverage 83%

Contents – 29 blocks, most important first

1. n_ 7

2. (bare) 2

3. comp_ 16

4. ctx_ 16

5. top_ 8

6. realized_ 5

7. w_ 1

8. lit_ 7

9. cptac_ 24

10. dominant_ 4

11. frac_ 3

12. static_ 1

13. total_ 1

14. owner_ 1
15. median_ 1

16. armres_ 7

17. identity_ 1

18. heur_ 5

19. greal_ 10

20. acquired_ 2

21. pressure_ 1

22. realization_ 1

23. tcga_ 4

24. disc_ 2

25. max_ 1

26. regulatory_ 1

27. mean_ 1

28. concentration_ 1

29. fam_ 1

N_ – 7

core – the columns findings rest on

n_arms GENEAGG ARM100%	Arms in the LEARNED MODEL's design – <code>x.shape[1]</code> from <code>assemble_gene</code> , carried via <code>gene_atlas</code> . This is the width the Gibbs actually fit (median 2, max 90).
n_cell_intrinsic GENE100%	FAMILIES whose coupling SURVIVES the composition block (retention – 0.7).
n_composition_explained GENE100%	FAMILIES whose coupling does NOT survive the composition block (retention < 0.4).
n_fam GENEAGG FAMILY100%	How many seed families regulate this gene.
n_hallmark_sets GENE91%	How many MSigDB Hallmark programs contain this gene (1–10, median 2). Membership only – it says nothing about regulation.
n_identified GENE100%	Families reaching $ z > 2$ on the UNCALIBRATED SD. Zero for 691 genes (44.6%) – meaning the model cannot distinguish ANY of that gene's families from zero.
n_pip_disc_gt50 GENE100%	Count of the gene's families with <code>pip_discovery > 0.5</code> under the evidence- $\tilde{\epsilon}$ readout.

(BARE) 2

core the columns findings rest on

concentration GENE 100%	<code>beta.max() / beta.sum()</code> the TOP FAMILY's share of the gene's total $\hat{\mu}^2$; median 0.954 , i.e. most genes are dominated by one family.
gene KEY 100%	KEY HGNC gene symbol.

COMP_ 16

core the columns findings rest on

comp_cptac_prot_driver GENE 59%	Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
comp_cptac_prot_driver_ret GENE 59%	<code>1 - comp_cptac_prot_driver_share</code> , exactly. Median +0.398 raw, +0.469 gated to $ \tilde{I}_{\text{raw}} \leq 0.10$ quote the gated value. See the mRNA twin. <i>Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)</i>
comp_cptac_prot_driver_share GENE 59%	Protein-layer twin of <code>comp_tcga_mrna_driver_share</code> . Same exact complementarity with <code>_driver_ret</code> (8.9e-16 on 918 rows).
comp_cptac_prot_rho_raw GENE 73%	Unadjusted aggregate PROTEIN coupling denominator of <code>comp_cptac_prot_driver_ret</code> .
comp_cptac_prot_top_budget GENE 73%	Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
comp_cptac_prot_top_budget_r GENE 73%	Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
comp_cptac_prot_top_target GENE 73%	Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
comp_cptac_prot_top_target_r GENE 73%	Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
comp_tcga_mrna_driver GENE 66%	WHICH compartment drives the gene's mRNA coupling the one whose removal moves it most (CAFs 279 · purity 174 · prolifer 148 · Endothelial 122 · Myeloid 109).
comp_tcga_mrna_driver_ret GENE 66%	<code>1 - comp_tcga_mrna_driver_share</code> , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment.
comp_tcga_mrna_driver_share GENE 66%	Share of the gene's mRNA coupling attributable to the single compartment that most changes it axiom 8's composition GATE, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10.
comp_tcga_mrna_rho_raw GENE 90%	The gene's UNADJUSTED aggregate mRNA coupling the DENOMINATOR of <code>comp_tcga_mrna_driver_ret</code> .
comp_tcga_mrna_top_budget GENE 90%	The compartment the gene's miRNA BUDGET loads on most a property of the REGULATORS.

comp_tcga_mrna_top_budget_r GENE 90%	Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
comp_tcga_mrna_top_target GENE 90%	The compartment the TARGET GENE's expression loads on most.
comp_tcga_mrna_top_target_r GENE 90%	Cell-composition drivers of this gene's coupling (<code>compartment_*</code>): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.

CTX_ Å· 16

core â€” the columns findings rest on

ctx_apriori_class GENE 100%	A-priori competence class from the gene atlas (A_COMPETENT / B_PARTIAL / C_WEAK / D_UNDEFINED).
ctx_ceiling GENE 98%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_collapse GENE 90%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_d_collin GENE 45%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_d_fam GENE 87%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_dose_max GENE 90%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_dose_med GENE 90%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_frac_abund GENE 90%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_gap_core GENE 87%	Real-minus-decoy coupling gap for this gene. Pooled it is ~-0.012, but the per-gene IQR STRADDLES ZERO â€” this is a set-level distributional shift, not a per-gene effect. Do not rank genes by it.
ctx_gap_d_dose GENE 87%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_gap_deconv GENE 87%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

ctx_gap_retention GENE 83%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_measurable GENE 98%	Is the gene's coupling measurable at all given its design (positive OOF ceiling). False for 877/1549 genes.
ctx_n_abund GENE 90%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_n_fam_abund GENE 90%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_n_fam_multi GENE 90%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

TOP_ 8

supporting “ read these when the core raises a question

top_beta GENE AGG FAMILY 100%	The gene's leading regulator and its value, by magnitude or by identity.
top_beta_frac GENE AGG FAMILY 100%	The gene's leading regulator and its value, by magnitude or by identity.
top_family_identity GENE AGG FAMILY 87%	The family Shapley/LMG credits with the gene's regulation. NaN for 197 genes (12.7%) where NNLS zeroed everything “ an honest 'undefined', which the magnitude answer cannot express.
top_family_magnitude GENE AGG FAMILY 100%	The family with the largest beta.
top_gaining_arms GENE 91%	The gene's leading regulator and its value, by magnitude or by identity.
top_identity GENE AGG FAMILY 87%	The largest <code>identity</code> share among the gene's families.
top_identity_gated GENE 78%	<code>top_identity</code> recomputed over <code>identity_reliable</code> edges only “ max 1.0000, 0 genes above 1 (vs +740.007 ungated). Defined on 1,203 genes.
top_identity_n_reliable GENE 78%	How many of the gene's families survive the identity gate “ the DENOMINATOR behind <code>top_identity_gated</code> . Median 2.

REALIZED_ 5

supporting “ read these when the core raises a question

realized_composition_explained GENE 82%	Realized (as opposed to potential) coupling for the gene “ the raw and adjusted rho, its retention, and how many regulators it was computed over.
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realized_n_reg GENE 82%	Realized (as opposed to potential) coupling for the gene “ the raw and adjusted rho, its retention, and how many regulators it was computed over.
realized_retention GENE 63%	Realized-coupling retention, $\bar{I}_{\text{adjusted}} / \bar{I}_{\text{raw}}$.
realized_rho_adj GENE 81%	Realized (as opposed to potential) coupling for the gene “ the raw and adjusted rho, its retention, and how many regulators it was computed over.
realized_rho_raw GENE 81%	Realized (as opposed to potential) coupling for the gene “ the raw and adjusted rho, its retention, and how many regulators it was computed over.

W_ 1
supporting “ read these when the core raises a question

w_max GENE 100%	The MAXIMUM curated evidence weight w over the gene's regulators (<code>gene_atlas : nanmax(w)</code>), where w comes from the same PMID-deduped ledger that feeds <code>lit_</code> and <code>fame_</code> .
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LIT_ 7
supporting “ read these when the core raises a question

lit_agrees_identity GENE 21%	Does <code>identity</code> name the same family as the most-published one for this gene “ True on 100 of 323 (31%) .
lit_agrees_magnitude GENE 21%	“” A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH “ read the name with suspicion. <code>eval/lit_ground_truth.py</code> defines the 'canonical' family of a gene as the argmax of DISTINCT PubMed IDs carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded).
lit_family GENE 21%	The most-published seed family for this gene by distinct low-throughput functional PMIDs.
lit_margin GENE 21%	Gap between the top and second literature family, in PMIDs “ median 1.00 , i.e. the literature's 'winner' is usually decided by a SINGLE paper. “ gate on this before reading <code>lit_agrees_*</code> : a 1-PMID margin is a coin flip dressed as ground truth. <i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>
lit_n_lit_families GENE 21%	“” A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH “ read the name with suspicion. <code>eval/lit_ground_truth.py</code> defines the 'canonical' family of a gene as the argmax of DISTINCT PubMed IDs carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded).
lit_n_pmid GENE 21%	Distinct low-throughput functional PMIDs behind <code>lit_family</code> . Median 3 “ the 'ground truth' rests on a handful of papers per gene. <i>Defined on: genes with a versioned literature canonical family (329 of 1,549, sha256-stamped, MH-196)</i>
lit_tier GENE 21%	“” A STUDY-COUNT AXIS, NOT A BIOLOGICAL GROUND TRUTH “ read the name with suspicion. <code>eval/lit_ground_truth.py</code> defines the 'canonical' family of a gene as the argmax of DISTINCT PubMed IDs carrying low-throughput functional evidence (reporter / western / proteomics; miRTarBase-Weak and TarBase-Positive excluded).

cptac_prosp_abund_rho_disc GENE AGG ARM 72%	CPTAC protein channel.
cptac_prosp_abund_rho_prot GENE AGG ARM 73%	CPTAC protein channel.
cptac_prosp_abund_rho_rna GENE AGG ARM 74%	CPTAC protein channel.
cptac_prosp_agg_beats_abund_prot GENE 73%	Does the learned $\hat{f}^2$ -weighted aggregate couple more negatively to PROTEIN than a plain abundance sum, in the CPTAC prospective cohort?
cptac_prosp_agg_ret_prot GENE 59%	rho_prot(composition-adjusted) / rho_prot_raw “ the ADJUSTMENT sense of retention, on the prospective cohort. “... The denominator is ALREADY GATED at <code>RHO_GATE</code> in <code>cptac_card.py:322</code> , so values outside [0,1] are NOT the vanishing-denominator artifact of axiom 5 (verified: 0 rows with <code> raw < 0.02</code>).
cptac_prosp_agg_rho_disc GENE AGG ARM 72%	CPTAC protein channel.
cptac_prosp_agg_rho_disc_raw GENE 72%	CPTAC protein channel.
cptac_prosp_agg_rho_prot GENE AGG ARM 73%	CPTAC protein channel.
cptac_prosp_agg_rho_prot_raw GENE 73%	CPTAC protein channel.
cptac_prosp_agg_rho_rna GENE AGG ARM 74%	CPTAC protein channel.
cptac_prosp_agg_rho_rna_raw GENE 74%	CPTAC protein channel.
cptac_prosp_n_arms GENE 89%	Arms contributing to this gene's prospective-cohort aggregate (median 2, max 61).
cptac_t105_abund_rho_disc GENE 60%	CPTAC protein channel.
cptac_t105_abund_rho_prot GENE 60%	CPTAC protein channel.
cptac_t105_abund_rho_rna GENE 60%	CPTAC protein channel.
cptac_t105_agg_beats_abund_prot GENE 60%	As the prospective twin, on the TCGA-105 overlap cohort.

cptac_t105_agg_ret_prot GENE42%	As the prospective twin, TCGA-105.
cptac_t105_agg_rho_disc GENEAGG ARM60%	CPTAC protein channel.
cptac_t105_agg_rho_disc_raw GENE60%	CPTAC protein channel.
cptac_t105_agg_rho_prot GENEAGG ARM60%	CPTAC protein channel.
cptac_t105_agg_rho_prot_raw GENE60%	CPTAC protein channel.
cptac_t105_agg_rho_rna GENEAGG ARM60%	CPTAC protein channel.
cptac_t105_agg_rho_rna_raw GENE60%	CPTAC protein channel.
cptac_t105_n_arms GENE85%	Arms behind the TCGA-105 aggregate (median 2, max 57). See the prospective twin. Defined on: genes/edges present in the CPTAC cohort

DOMINANT_ 4

reference context, provenance and rarely-quoted detail

dominant_HLY GENE77%	The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to <code>regulatory_handoff</code> .
dominant_NAT GENE81%	The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to <code>regulatory_handoff</code> .
dominant_TUM GENE82%	The arm carrying the largest share of the gene's tumour pressure budget (miR-21-5p on 56 genes, miR-200c-3p next).
dominant_edge_shift_class GENE83%	The top-ranked regulator of this gene in one state (HLY / NAT / TUM), by GLOBAL dose rank. The inputs to <code>regulatory_handoff</code> .

FRAC_ 3

reference context, provenance and rarely-quoted detail

frac_edges_realized GENE82%	A fraction read the suffix for the numerator and <code>domain</code> for the denominator's rows.
frac_gain_true_acquired GENE60%	A fraction read the suffix for the numerator and <code>domain</code> for the denominator's rows. Defined on: genes with a NAT-TUM gene-level leg
frac_identified GENE100%	A fraction read the suffix for the numerator and <code>domain</code> for the denominator's rows.

STATIC_ 1

reference “ context, provenance and rarely-quoted detail

static_owner_family

GENE

87%

The static (dose-based) owner of the gene, for comparison against the realization owner.

Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

TOTAL_ 1

reference “ context, provenance and rarely-quoted detail

total_pressure

GENE

100%

Sum over the gene's regulators.

OWNER_ 1

reference “ context, provenance and rarely-quoted detail

owner_agrees

GENE

41%

Whether the realization owner and the static owner agree. Defined only where both exist.

Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)

MEDIAN_ 1

reference “ context, provenance and rarely-quoted detail

median_retention

GENE

100%

A per-row median of a lower-rung quantity; check `agg_of`.

ARMRES_ 7

reference “ context, provenance and rarely-quoted detail

armres_best_drho

GENE

18%

“ IS THIS GENE MODELABLE AT ARM RESOLUTION “ the gene-level roll-up of arm identifiability (`card_ladders.gene_arm_resolution`, 2026-08-19).

armres_class

GENE

18%

`arm_modelable` (every multi-arm cell resolvable, 68 genes) “ `partial` (37) “ `family_only` (no cell resolvable, 168). Absent = the gene has no multi-arm cell at all, so the question is undefined.

armres_frac_resolvable

GENE

18%

Fraction of this gene's multi-arm cells where the arms are separable.

armres_med_drho

GENE

18%

Median `oof_drho` over the gene's multi-arm cells “ negative means arm resolution predicts repression better out of fold.

armres_med_sep_z

GENE

18%

“ IS THIS GENE MODELABLE AT ARM RESOLUTION “ the gene-level roll-up of arm identifiability (`card_ladders.gene_arm_resolution`, 2026-08-19).

armres_n_arms_split

GENE

18%

“ IS THIS GENE MODELABLE AT ARM RESOLUTION “ the gene-level roll-up of arm identifiability (`card_ladders.gene_arm_resolution`, 2026-08-19).

armres_n_multi_cells

GENE

18%

“ IS THIS GENE MODELABLE AT ARM RESOLUTION “ the gene-level roll-up of arm identifiability (`card_ladders.gene_arm_resolution`, 2026-08-19).

IDENTITY_ 1

reference " context, provenance and rarely-quoted detail

identity_eq_magnitude	Do the identity and magnitude answers name the SAME top family. Exactly 100% agreement at n_fam==1 by construction; 76% at n_fam>=2. Read it only on multi-family genes.
GENE	AGG FAMILY 87%

HEUR_ 5

reference " context, provenance and rarely-quoted detail

heur_delta_tumor_nat	Tumour-minus-NAT change in the heuristic pressure correlation.
GENE	89%
heur_n_regulators	A DIFFERENT PIPELINE'S COUNT " not the model's. It is edges.groupby('gene')['miRNA'].nunique() inside analyses/misc/mirna_comovement.py , i.e.
GENE	91%
heur_net_repressed_tumor	Is the gene net-repressed in tumour, per the RETIRED heuristic lane. True for 282 of 1,409 genes.
GENE	91%
heur_repression_class	Cross-state repression class from the RETIRED heuristic lane (never_repressed / lost_repression / gained_repression / constitutive_repressed). 867 of 1,409 genes are never_repressed .
GENE	91%
heur_rho_pressure_tumor	Spearman of the gene's HEURISTIC aggregate pressure against its mRNA in tumour.
GENE	91%

GREAL_ 10

reference " context, provenance and rarely-quoted detail

greal_best_coupling	Realization ladder for the GENE " the mirror of real_ : how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). æ... Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) " the GENE-side ladder; rates gated n3
GENE	83%
greal_frac_c10	Gene-rung twin of real_frac_c10 " defined on 446 genes against greal_n_c10 's 1,289; gated-out genes have a median greal_n_scored of 1. Exact on 100%. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) " the GENE-side ladder; rates gated n3
GENE	29%
greal_med_coupling	Realization ladder for the GENE " the mirror of real_ : how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). æ... Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) " the GENE-side ladder; rates gated n3
GENE	83%
greal_n_c05	Realization ladder for the GENE " the mirror of real_ : how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). æ... Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) " the GENE-side ladder; rates gated n3
GENE	83%
greal_n_c10	Realization ladder for the GENE " the mirror of real_ : how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). æ... Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) " the GENE-side ladder; rates gated n3
GENE	83%

greal_n_c15 GENE 83%	Realization ladder for the GENE – the mirror of <code>real_</code>: how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). – Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) – the GENE-side ladder; rates gated n3
greal_n_c20 GENE 83%	Realization ladder for the GENE – the mirror of <code>real_</code>: how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). – Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) – the GENE-side ladder; rates gated n3
greal_n_c30 GENE 83%	Realization ladder for the GENE – the mirror of <code>real_</code>: how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). – Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) – the GENE-side ladder; rates gated n3
greal_n_scored GENE 83%	Genes/edges scored for realized coupling (median 2, max 59) – the DENOMINATOR of every <code>greal_frac_*</code>. – The <code>greal_n_c05</code> <code>n_c30</code> ladder is properly nested: 0 monotonicity violations and 0 rows where a threshold count exceeds <code>n_scored</code> (verified). Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) – the GENE-side ladder; rates gated n3
greal_n_sig GENE 83%	Realization ladder for the GENE – the mirror of <code>real_</code>: how many of THIS gene's regulators realize at each calibrated-coupling depth (c05/c10/c15/c20/c30). – Cross-rung control: the gene-side totals reproduce the arm-side exactly. Defined on: genes with 1 calibrated-coupling-scored regulator (1,260) – the GENE-side ladder; rates gated n3

ACQUIRED_ 2

reference – context, provenance and rarely-quoted detail

acquired_vs_gtex GENE 91%	Acquisition of regulatory dose relative to a healthy reference (GTEx or NAT).
acquired_vs_nat GENE 91%	Acquisition of regulatory dose relative to a healthy reference (GTEx or NAT).

PRESSURE_ 1

reference – context, provenance and rarely-quoted detail

pressure_tumor GENE 91%	Aggregate incoming pressure on the gene in the named state.
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REALIZATION_ 1

reference – context, provenance and rarely-quoted detail

realization_owner GENE 42%	Within-patient realization quantities – the score, its identity allocation, and who owns it. Defined on: MULTI-FAMILY genes only (an owner needs >1 family to choose between)
---	--

TCGA_ 4

reference context, provenance and rarely-quoted detail

tcga_abund_rho_rna GENEAGG ARM89%	The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.
tcga_agg_rho_rna GENEAGG ARM89%	Gene-level $\hat{l}^2$ -weighted aggregate coupling to mRNA in TCGA (median $\hat{\alpha}^{*0.05}$) against <code>tcga_abund_rho_rna</code> 's $\hat{\alpha}^{*0.02}$ the abundance-sum reference.
tcga_agg_rho_rna_raw GENE89%	The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.
tcga_n_arms GENE89%	The TCGA mRNA leg carried beside the CPTAC columns so the two are comparable on the same gene rows.

DISC_ 2

reference context, provenance and rarely-quoted detail

disc_gold_families GENE3%	Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed).
disc_n_gold_edges GENE100%	How many of the 157 discovery-queue edges name this gene (max 4).

MAX_ 1

reference context, provenance and rarely-quoted detail

max_beta_frac_sd GENE100%	Maximum over the row's constituent units.
---	---

REGULATORY_ 1

reference context, provenance and rarely-quoted detail

regulatory_handoff GENE100%	Does the gene's globally-dominant regulator differ between healthy and tumour OR between NAT and tumour.
---	--

MEAN_ 1

reference context, provenance and rarely-quoted detail

mean_dTarget GENE82%	A per-row mean of a lower-rung quantity; check <code>agg_of</code> for which rung it summarises.
--	--

CONCENTRATION_ 1

reference context, provenance and rarely-quoted detail

concentration_adj GENE53%	$(\text{concentration} \hat{\alpha}^{*} 1/n_{fam}) / (1 \hat{\alpha}^{*} 1/n_{fam}) \hat{\alpha}^{*} [0,1]$ how concentrated the gene is RELATIVE to the most diffuse its own design allows.
---	--

FAM_ 1

reference context, provenance and rarely-quoted detail

fam_pooled_rho_adj

Seed-family structure the arm sits in membership, affinity and dominance within it.

GENE

81%

gene_family_card.tsv

gene_family

One row per one (seed family, gene) pair. Keyed [gene, family] it cannot express a property of the family itself. That is the seed_family card's job.

61 columns 5,117 rows 12 blocks median coverage 97%

Contents 12 blocks, most important first

1. (bare) 8 CORE

2. n_3 SUPPORTING

3. ctx_16 SUPPORTING

4. beta_7 SUPPORTING

5. identity_4 REFERENCE

6. comp_16 REFERENCE
7. w_1 REFERENCE

8. pip_2 REFERENCE

9. m_1 REFERENCE

10. composition_1 REFERENCE

11. prior_1 REFERENCE

12. retention_1 REFERENCE

(BARE) 8

core the columns findings rest on

arms FAMILY100%	The arms collapsed into this (gene, family) cell, semicolon-separated (hsa-miR-17-5p;hsa-miR-20a-5p).
beta FAMILY100%	FAMILY-rung coupling coefficient readouts.run(level='family') , the same-seed collapse APPLIED.
family KEY100%	KEY seed family of the regulator, as used by the family-rung fit.
gene KEY100%	KEY HGNC gene symbol.
identified FAMILY100%	z > 2 on the UNCALIBRATED SD 27.1% of cells. cal_identified is the honest version.
identity FAMILY95%	Shapley/LMG identity at the FAMILY rung same estimator as the edge card's identity , different unit.
retention FAMILY100%	beta_deconv / beta_core at the family grain the ADJUSTMENT sense of retention (what fraction of the coupling survives the composition block), never the patient-baseline sense.
z FAMILY100%	beta / beta_sd for the family cell, on the UNCALIBRATED posterior SD.

N_ 3

supporting read these when the core raises a question

n_arms FAMILY100%	How many arms of THIS family sit in THIS gene's design. ... FAMILY rung, VERIFIED (2026-08-19): it varies within gene for 241 of 1,549 genes, and the per-gene SUM of these equals the gene card's n_arms exactly a clean cross-rung consistency check.
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n_fam	How many families compete at this gene. beta is mathematically INERT at 1.
GENE 100%	
n_samples	Samples the family cell was fit on â€” CONSTANT 1,040 across the card.
GENE 100%	
CTX_ Â• 16 supporting â€” read these when the core raises a question	
ctx_apriori_class	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 100%	
ctx_ceiling	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 99%	
ctx_collapse	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 96%	
ctx_d_collin	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 81%	
ctx_d_fam	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 95%	
ctx_dose_max	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 96%	
ctx_dose_med	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 96%	
ctx_frac_abund	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 96%	
ctx_gap_core	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 95%	
ctx_gap_d_dose	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 95%	
ctx_gap_deconv	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 95%	
ctx_gap_retention	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
GENE 92%	

ctx_measurable GENE 99%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_n_abund GENE 96%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_n_fam_abund GENE 96%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_n_fam_multi GENE 96%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

BETA_ 7

supporting “ read these when the core raises a question

beta_deconv FAMILY 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta.
beta_frac FAMILY 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta.
beta_frac_abs FAMILY 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta.
beta_frac_deconv FAMILY 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta.
beta_frac_sd FAMILY 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta.
beta_sd FAMILY 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta.
beta_sd_deconv FAMILY 100%	Posterior summaries of the coupling coefficient from the dense Gibbs fit (<code>learned/readouts.py</code>). <code>_sd</code> is the posterior SD, <code>_frac</code> the share of the gene's total beta.

IDENTITY_ 4

reference “ context, provenance and rarely-quoted detail

identity_abs FAMILY 95%	$ identity / \hat{E} identity $ per gene “ the BOUNDED companion to <code>identity</code> , always in [0,1] (verified: 0 non-NaN values outside it on either card).
identity_coherence FAMILY 95%	$1 / \hat{E} identity $ per GENE, in (0,1] “ identity's own sign coherence. 1.0 when nothing cancels, “ 0 as suppressor cancellation grows.
identity_deconv FAMILY 97%	Shapley/LMG attribution on R^2 with bagged-NNLS weights “ WHO regulates the gene, collinearity-fair.
identity_reliable FAMILY 95%	<code>identity_coherence</code> “ 0.5 AND $ identity $ “ 1. Admits 93.3% of edge rows and 92.6% of gene_family rows, and excludes every row whose $ identity $ exceeds 1 (119/119 and 118/118).

COMP_ 16

reference “ context, provenance and rarely-quoted detail

comp_cptac_prot_driver	Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
GENE 66%	
comp_cptac_prot_driver_ret	1 ^ ' comp_cptac_prot_driver_share , exactly. Median +0.398 raw, +0.469 gated to ĩ_raw % ¥ 0.10 “ quote the gated value. See the mRNA twin.
GENE 66%	Defined on: genes in the CPTAC-PROTEIN compartment analysis (protein layer, n~101)
comp_cptac_prot_driver_share	Protein-layer twin of comp_tcga_mrna_driver_share . Same exact complementarity with _driver_ret (8.9e-16 on 918 rows).
GENE 66%	
comp_cptac_prot_rho_raw	Unadjusted aggregate PROTEIN coupling “ denominator of comp_cptac_prot_driver_ret .
GENE 81%	
comp_cptac_prot_top_budget	Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
GENE 81%	
comp_cptac_prot_top_budget_r	Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
GENE 81%	
comp_cptac_prot_top_target	Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
GENE 81%	
comp_cptac_prot_top_target_r	Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
GENE 81%	
comp_tcga_mrna_driver	WHICH compartment drives the gene's mRNA coupling “ the one whose removal moves it most (CAFs 279 Â· purity 174 Â· prolif 148 Â· Endothelial 122 Â· Myeloid 109).
GENE 81%	
comp_tcga_mrna_driver_ret	1 ^ ' comp_tcga_mrna_driver_share , exactly (see there). The RETENTION orientation: what fraction of the coupling survives removing the driver compartment.
GENE 81%	
comp_tcga_mrna_driver_share	Share of the gene's mRNA coupling attributable to the single compartment that most changes it “ axiom 8's composition GATE, the axis that separates the harm tails at p=0.00999 where the ensemble axis gives p=0.10.
GENE 81%	
comp_tcga_mrna_rho_raw	The gene's UNADJUSTED aggregate mRNA coupling “ the DENOMINATOR of comp_tcga_mrna_driver_ret .
GENE 97%	
comp_tcga_mrna_top_budget	The compartment the gene's miRNA BUDGET loads on most “ a property of the REGULATORS.
GENE 97%	
comp_tcga_mrna_top_budget_r	Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
GENE 97%	
comp_tcga_mrna_top_target	The compartment the TARGET GENE's expression loads on most.
GENE 97%	
comp_tcga_mrna_top_target_r	Cell-composition drivers of this gene's coupling (compartment_*): which deconvolved fraction explains the correlation, its retention under adjustment, and its share.
GENE 97%	

W_ 1

reference " context, provenance and rarely-quoted detail

w_max	Maximum curated EVIDENCE weight for this (gene, family) cell. Family-rung here; gene-rung on the gene card.
GENE100%	

PIP_ 2

reference " context, provenance and rarely-quoted detail

pip_dense	CONSTANT by construction (i=1 dense readout).
FAMILY100%	
pip_discovery	Posterior inclusion probability. pip_dense comes from the pi=1 coupling readout; pip_discovery from the evidence-pi readout.
FAMILY100%	

M_ 1

reference " context, provenance and rarely-quoted detail

m_nnl	Bagged-NNLS weight for this (gene, family) cell.
FAMILY100%	

COMPOSITION_ 1

reference " context, provenance and rarely-quoted detail

composition_class	Composition-adjustment verdict for this row (e.g. composition_explained when the effect does not survive the deconvolution block).
FAMILY100%	

PRIOR_ 1

reference " context, provenance and rarely-quoted detail

prior_pi	The evidence prior fed to the discovery readout.
FAMILY100%	

RETENTION_ 1

reference " context, provenance and rarely-quoted detail

retention_reliable	Fraction of an effect surviving covariate adjustment (adjusted / raw).
FAMILY100%	

arm

296 columns **2,450** rows **43** blocks median coverage **30%**

```

1. adm_ 4 Â· CORE
2. arm_ 7 Â· CORE
3. seed_ 1 Â· CORE
4. site_ 12 Â· SUPPORTING
5. arb_ 16 Â· SUPPORTING
6. healthy_ 3 Â· SUPPORTING
7. (bare) 4 Â· SUPPORTING
8. fame_ 28 Â· SUPPORTING
9. ago_ 4 Â· SUPPORTING
10. hly_ 7 Â· SUPPORTING
11. dose_ 3 Â· SUPPORTING
12. abund_ 5 Â· SUPPORTING
13. attr_ 14 Â· REFERENCE
14. ctx_ 2 Â· REFERENCE
15. grank_ 3 Â· REFERENCE
16. real_ 11 Â· REFERENCE
17. ts_ 10 Â· REFERENCE
18. isoc_ 5 Â· REFERENCE
19. sdsz_ 4 Â· REFERENCE
20. fst_ 16 Â· REFERENCE
21. chim_ 5 Â· REFERENCE
22. field_ 5 Â· REFERENCE
23. sub_ 6 Â· REFERENCE
24. seq_ 4 Â· REFERENCE
25. cov_ 22 Â· REFERENCE
26. cur_ 2 Â· REFERENCE
27. iso_ 5 Â· REFERENCE
28. famrole_ 6 Â· REFERENCE
29. surv_ 8 Â· REFERENCE
30. aid_ 4 Â· REFERENCE
31. bc_ 11 Â· REFERENCE
32. wshift_ 6 Â· REFERENCE
33. tier_ 5 Â· REFERENCE
34. disc_ 2 Â· REFERENCE
35. cnv_ 9 Â· REFERENCE
36. gctx_ 16 Â· REFERENCE
37. surrogate_ 2 Â· REFERENCE
38. cnvc_ 5 Â· REFERENCE
39. foc_ 7 Â· REFERENCE
40. model_ 1 Â· REFERENCE
41. comv_ 2 Â· REFERENCE
42. cload_ 3 Â· REFERENCE
43. hx_ 1 Â· REFERENCE

```

core “ the columns findings rest on

ARM 14%

ARM 29%

ARM 29%

Defined on: edges/arms with an admissibility tag (4,937 edges; has_site 89.9% Â· expressed 63.9% Â· admissible 60.4%)

ARM_ 7

core the columns findings rest on

arm_iqr ARM20%	Interquartile range of the arm's tumour RPM its DYNAMIC RANGE.
arm_lfc_HLY_NAT_raw ARM15%	log-FC GTEx-healthy TCGA-NAT, raw.
arm_lfc_HLY_TUM_QN ARM16%	log-FC GTEx-healthy TCGA-tumour, on quantile-normalised values.
arm_lfc_HLY_TUM_raw ARM15%	As arm_lfc_HLY_TUM_QN but against the RAW GTEx median.
arm_lfc_NAT_TUM ARM20%	log-FC of the arm's median abundance, matched NAT tumour.
arm_med_rpm ARM20%	The arm's MEDIAN RPM across tumour samples its abundance level, gene-free. Defined on the 19.8% of arms with tumour expression data. Defined on: edges present in the progression panel (arm expressed + state measured)
arm_pct_floor ARM	THE NAME READS BACKWARDS. Computed as $100 \cdot \text{mean}(x > \text{FL00R})$ the percent of samples ABOVE the detection floor, i.e.

SEED_ 1

core the columns findings rest on

seed_family ARM100%	KEY the seed family itself, one row per family.
---	---

SITE_ 12

supporting read these when the core raises a question

site_frac_8mer ARM29%	8mer share of the arm's CANONICAL sites ($\text{site_n_8mer} / \text{site_n_canonical}$; median 0.150).
site_frac_canonical ARM29%	Canonical share of the arm's TOTAL sites ($\text{site_n_canonical} / \text{site_n_total}$; median 0.054) a different denominator from site_frac_8mer.
site_n_6mer ARMAGG EDGE29%	Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.
site_n_7mer ARMAGG EDGE29%	Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.
site_n_8mer ARMAGG EDGE29%	Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.
site_n_canonical ARMAGG EDGE29%	Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.
site_n_genes_canonical ARMAGG EDGE29%	Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.

site_n_noncanon ARM AGG EDGE 29%	Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.
site_n_total ARM AGG EDGE 29%	Seed-site counts in the target 3'UTR by site type (8mer / 7mer / 6mer / non-canonical) from the genome-wide scan.
site_repression_med ARM AGG EDGE 29%	Median predicted repression score over the arm's sites (all negative by construction; median $\hat{\sim}0.316$). A PREDICTION from sequence, not a measurement $\hat{\sim}$ never cite it as evidence of observed repression. Defined on: arms in the scanMiR site-type table (771) $\hat{\sim}$ HALLMARK-SCOPED, 1,432 genes only
site_repression_min ARM AGG EDGE 29%	Strongest (most negative) predicted repression among the arm's sites (median $\hat{\sim}1.964$).
site_sites_per_gene ARM 29%	Canonical sites per targeted gene (median 1.34) $\hat{\sim}$ site DENSITY, separating an arm with many sites on few genes from one spread thin. <code>site_n_canonical / site_n_genes_canonical</code> . Defined on: arms in the scanMiR site-type table (771) $\hat{\sim}$ HALLMARK-SCOPED, 1,432 genes only

ARB_ 16

supporting $\hat{\sim}$ read these when the core raises a question

arb_frac_identified ARM AGG EDGE 30%	<code>arb_n_identified / arb_n_edges</code> (verified exact on 100% of rows).
arb_max_abs_beta ARM AGG EDGE 30%	Largest $ \hat{\sim} ^2$ among the arm's edges.
arb_max_identity ARM AGG EDGE 28%	Largest <code>identity</code> among this arm's edges, over reliable rows only. $\hat{\sim}$... Now runs $\hat{\sim}0.255$ $\hat{\sim}$ +1.000 $\hat{\sim}$ it reached +740.0 until the inert <code>beta_frac_reliable</code> gate was replaced (2026-08-19).
arb_mean_abs_beta ARM AGG EDGE 30%	Mean $ \hat{\sim} ^2$ over the arm's edges. See <code>arb_max_abs_beta</code> for the one-edge degeneracy (9.3% of arms, where the two are equal by construction). Defined on: arms with a learned-$\hat{\sim}^2$ edge rollup (728)
arb_n_b001 ARM 30%	Per-ARM rollup of its edge set $\hat{\sim}$ how many edges and genes it regulates and its mean $ \text{beta} $ over them.
arb_n_b001_ident ARM 30%	Per-ARM rollup of its edge set $\hat{\sim}$ how many edges and genes it regulates and its mean $ \text{beta} $ over them.
arb_n_b005 ARM 30%	Per-ARM rollup of its edge set $\hat{\sim}$ how many edges and genes it regulates and its mean $ \text{beta} $ over them.
arb_n_b005_ident ARM 30%	Per-ARM rollup of its edge set $\hat{\sim}$ how many edges and genes it regulates and its mean $ \text{beta} $ over them.
arb_n_b010 ARM 30%	Per-ARM rollup of its edge set $\hat{\sim}$ how many edges and genes it regulates and its mean $ \text{beta} $ over them.
arb_n_b010_ident ARM 30%	Per-ARM rollup of its edge set $\hat{\sim}$ how many edges and genes it regulates and its mean $ \text{beta} $ over them.
arb_n_b025 ARM 30%	Per-ARM rollup of its edge set $\hat{\sim}$ how many edges and genes it regulates and its mean $ \text{beta} $ over them.

arb_n_b025_ident ARM 30%	Per-ARM rollup of its edge set â€” how many edges and genes it regulates and its mean beta over them.
arb_n_comp_explained ARM AGG EDGE 30%	Per-ARM rollup of its edge set â€” how many edges and genes it regulates and its mean beta over them.
arb_n_edges ARM AGG EDGE 30%	Edges this arm carries in the LEARNED MODEL's own readouts.
arb_n_identified ARM AGG EDGE 30%	Per-ARM rollup of its edge set â€” how many edges and genes it regulates and its mean beta over them.
arb_n_identity_reliable ARM AGG EDGE 30%	Edges of this arm whose <code>identity</code> share is usable (<code>identity_reliable</code>).

HEALTHY_ 3

supporting â€” read these when the core raises a question

healthy_leg ARM 20%	Healthy-state legs and flags â€” including <code>healthy_uninformative</code> , which marks rows where the healthy baseline cannot carry a claim. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)
healthy_potential ARM 16%	Healthy-state legs and flags â€” including <code>healthy_uninformative</code> , which marks rows where the healthy baseline cannot carry a claim. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)
healthy_uninformative ARM 16%	Healthy-state legs and flags â€” including <code>healthy_uninformative</code> , which marks rows where the healthy baseline cannot carry a claim. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)

(BARE) 4

supporting â€” read these when the core raises a question

arm KEY 100%	KEY â€” mature miRNA arm (e.g. hsa-miR-21-5p). Normalised through <code>_arm_key()</code> , which gates against the canonical universe and LOGS drops rather than silently dropping.
dGlobal_HLY_NAT ARM 15%	Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)
dGlobal_HLY_TUM ARM 15%	Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state â€” many arms are multi-mapping-collapsed, MH-166)
dGlobal_NAT_TUM ARM 20%	Change in the arm's GLOBAL (cohort-wide) dose rank between two states. HLY_NAT is the FIELD delta, NAT_TUM the malignant step. Defined on: edges with a matched NAT leg (n~104 paired participants)

FAME_ 28

supporting “ read these when the core raises a question

fame_assay_binding_functional_mti_weak_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_binding_studies ARM 96%	Studies whose assay class is BINDING (the marginal of the assay – functional-class cross-tab).
fame_assay_functional_mti_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_functional_mti_weak_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_nonfunctional_mti_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_other_functional_mti_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_other_functional_mti_weak_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_other_nonfunctional_mti_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_other_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_protein_functional_mti_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_protein_functional_mti_weak_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_protein_nonfunctional_mti_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_protein_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.
fame_assay_reporter_functional_mti_studies ARM 96%	Study-attention axis for the arm “ distinct PMIDs and distinct curated genes.

fame_assay_reporter__nonfunctional_mti_studies	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM 96%	
fame_assay_reporter_studies	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM 96%	
fame_assay_rna__functional_mti_studies	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM 96%	
fame_assay_rna__functional_mti_weak_studies	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM 96%	
fame_assay_rna__nonfunctional_mti_studies	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM 96%	
fame_assay_rna_studies	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM 96%	
fame_assay_total_studies	Distinct STUDIES (<code>study_id.nunique()</code>) behind this arm's curated edges â€™ median 57, max 930. See <code>fame_assay_unique_targets</code> : different definitions, one axis (spearman 1.0000).
ARM 96%	
fame_assay_unique_targets	Distinct TARGET GENES this arm has in the curated literature (<code>gene.nunique()</code>).
ARM 96%	
fame_led_frac_ltfunc	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM AGG EDGE 96%	
fame_led_n_classes	How many distinct evidence CLASSES this arm's curated edges span (1â€™7).
ARM AGG EDGE 96%	
fame_led_n_ltfunc_pmid	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM AGG EDGE 96%	
fame_led_n_weak	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM AGG EDGE 94%	
fame_n_genes_curated	Study-attention axis for the arm â€™ distinct PMIDs and distinct curated genes.
ARM AGG EDGE 96%	
fame_npmid	Distinct PMIDs citing this arm.
ARM AGG EDGE 96%	

AGO_ 4

supporting â€™ read these when the core raises a question

ago_dom	Fraction of the arm's Manakov AGO-CLIP reads assigned to it rather than its sibling arm â€™ the loading dominance. Median 0.658 on the 27.7% of arms with CLIP coverage.
ARM 28%	
ago_dom_src	Provenance of <code>ago_dom</code> â€™ which CLIP dataset the AGO-loading dominance came from.
ARM 28%	

ago_guide_class	AGO/RISC loading gate for the arm.
ARM28%	
ago_reads	Raw AGO-CLIP read count (median 6, max 210,722).
ARM60%	

HLY_ 7
supporting read these when the core raises a question

hly_baseline_src	HEALTHY baseline levels for the arm GTEEx breast median/percentile and TCGA-NAT median. <i>Defined on: arms with a GTEEx healthy anchor MEASURED leg only; floor0 is NEVER emitted as a value</i>
ARM91%	
hly_from_seedmate	HEALTHY baseline levels for the arm GTEEx breast median/percentile and TCGA-NAT median. <i>Defined on: arms with a GTEEx healthy anchor MEASURED leg only; floor0 is NEVER emitted as a value</i>
ARM91%	
hly_gtex_median	HEALTHY baseline levels for the arm GTEEx breast median/percentile and TCGA-NAT median. <i>Defined on: arms with a GTEEx healthy anchor MEASURED leg only; floor0 is NEVER emitted as a value</i>
ARM17%	
hly_gtex_pct	HEALTHY baseline levels for the arm GTEEx breast median/percentile and TCGA-NAT median. <i>Defined on: arms with a GTEEx healthy anchor MEASURED leg only; floor0 is NEVER emitted as a value</i>
ARM17%	
hly_nat_detect_n	In how many of the ~104 matched NAT samples this arm is detected above the floor.
ARM91%	
hly_nat_median	Median NAT abundance (fill 74.0%).
ARM74%	
hly_tumor_prev	HEALTHY baseline levels for the arm GTEEx breast median/percentile and TCGA-NAT median. <i>Defined on: arms with a GTEEx healthy anchor MEASURED leg only; floor0 is NEVER emitted as a value</i>
ARM91%	

DOSE_ 3
supporting read these when the core raises a question

dose_comp_retention	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. <i>Defined on: edges with a matched NAT leg (n~104 paired participants)</i>
ARM12%	
dose_confounded	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. <i>Defined on: edges with a matched NAT leg (n~104 paired participants)</i>
ARM12%	
dose_prolif_retention	Dose (abundance) of the regulator and how it shifts across states, plus host-gene co-transcription fractions where the arm is intragenic. <i>Defined on: edges with a matched NAT leg (n~104 paired participants)</i>
ARM12%	

ABUND_ 5

supporting “ read these when the core raises a question

abund_frac_detected ARM 91%	The arm's own abundance profile across the cohort “ median, SD, IQR, detection fraction.
abund_iqr ARM 91%	The arm's own abundance profile across the cohort “ median, SD, IQR, detection fraction.
abund_med ARM 91%	The arm's own abundance profile across the cohort “ median, SD, IQR, detection fraction.
abund_sd ARM 88%	The arm's own abundance profile across the cohort “ median, SD, IQR, detection fraction.
abund_sd_tertile ARM 88%	Dispersion tertile “ 718 / 716 / 718 by construction , so it carries no information beyond <code>abund_sd</code> 's rank.

ATTR_ 14

reference “ context, provenance and rarely-quoted detail

attr_max_budget_share ARM 15%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>
attr_max_credit_share ARM 6%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>
attr_max_joint_share ARM 15%	Largest JOINT attribution share among this arm's edges.
attr_max_nested_share ARM 15%	Largest NESTED attribution share. See <code>attr_max_joint_share</code> : different values (agree on 35.4%), one axis (spearman +0.962). <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>
attr_max_within_share ARM 15%	Largest WITHIN-cell attribution share for this arm.
attr_med_budget_share ARM 15%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>
attr_med_credit_share ARM 6%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>
attr_med_dose_share ARM 29%	Median dose share of this arm within its family cells.
attr_med_joint_share ARM 15%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>
attr_med_nested_share ARM 15%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. <i>Defined on: arms with a Shapley/attribution decomposition over their edges (728)</i>

attr_med_phi_joint ARM 15%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_med_within_share ARM 15%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_n_edges ARM 22%	Attribution rollup over the arm's edges “ how much of the gene's coupling this arm is credited with, and over how many reliable edges. Defined on: arms with a Shapley/attribution decomposition over their edges (728)
attr_n_reliable ARM 15%	Reliable edges behind this arm's attr_* shares (median 2).

CTX_ 2

reference “ context, provenance and rarely-quoted detail

ctx_arm_abundant ARM 29%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.
ctx_arm_dose ARM 29%	Gene-design context from the decoy/atlas layer (<code>learned/analyses/gene_atlas.py</code> via <code>card_context</code>): how wide and how measurable this gene's regulator design is, and the real-vs-decoy gap measured on it.

GRANK_ 3

reference “ context, provenance and rarely-quoted detail

grank_HLY ARM 15%	The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from. Defined on: arms with a GTEx-HEALTHY leg (the thinnest state “ many arms are multi-mapping-collapsed, MH-166)
grank_NAT ARM 20%	The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from. Defined on: edges with a matched NAT leg (n~104 paired participants)
grank_TUM ARM 20%	The arm's GLOBAL (cohort-wide) dose rank in one state. This is what the dominant-regulator and handoff columns are computed from. Defined on: edges present in the progression panel (arm expressed + state measured)

REAL_ 11

reference “ context, provenance and rarely-quoted detail

real_best_coupling ARM 24%	Most negative realized coupling among this arm's scored genes (median ^'0.08, min ^'0.56).
real_frac_c10 ARM 9%	Fraction of this arm's scored genes coupling below ^'0.10.

real_frac_sig ARM9%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_med_coupling ARM24%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_n_c05 ARM24%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_n_c10 ARM24%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_n_c15 ARM24%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_n_c20 ARM24%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_n_c30 ARM24%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_n_scored ARM24%	Genes scored for this arm's realized coupling (median 3). œ... Its n_c05 “ n_c30 ladder is properly nested “ 0 violations (verified). <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>
real_n_sig ARM24%	Realization rollup for the ARM “ how many of its edges are scored, and its median and best realized coupling. <i>Defined on: arms with 1 calibrated-coupling-scored edge (593) “ the realization LADDER; rates are gated at n5</i>

TS_ Å· 10
reference “ context, provenance and rarely-quoted detail

ts_ctx_best ARM13%	Best (most negative) TargetScan context++ score for this arm “ range [â`2.02, â`0.34], all negative by construction (context++ scores repression as a negative number).
ts_ctx_mean ARM13%	TargetScan context++ scores for the arm's sites “ number of sites, mean and best context score. More negative context++ = stronger predicted repression. <i>Defined on: arms in TargetScan's human default predictions (âš only 321)</i>
ts_frac_8mer ARM13%	TargetScan context++ scores for the arm's sites “ number of sites, mean and best context score. More negative context++ = stronger predicted repression. <i>Defined on: arms in TargetScan's human default predictions (âš only 321)</i>

ts_kd_med ARM13%	TargetScan context++ scores for the arm's sites â€” number of sites, mean and best context score. More negative context++ = stronger predicted repression. Defined on: arms in TargetScan's human default predictions (âš only 321)
ts_n_7mer_A1 ARM13%	TargetScan context++ scores for the arm's sites â€” number of sites, mean and best context score. More negative context++ = stronger predicted repression. Defined on: arms in TargetScan's human default predictions (âš only 321)
ts_n_7mer_m8 ARM13%	TargetScan context++ scores for the arm's sites â€” number of sites, mean and best context score. More negative context++ = stronger predicted repression. Defined on: arms in TargetScan's human default predictions (âš only 321)
ts_n_8mer ARM13%	TargetScan context++ scores for the arm's sites â€” number of sites, mean and best context score. More negative context++ = stronger predicted repression. Defined on: arms in TargetScan's human default predictions (âš only 321)
ts_n_genes ARMAGG EDGE13%	Genes with a TargetScan context++ site â€” median 497, fill 12.9%, the SPARSEST universe on the card . Correlates +0.726 with the MANE seed scan and only +0.231 with scanMiR-any.
ts_n_sites ARMAGG EDGE13%	TargetScan context++ scores for the arm's sites â€” number of sites, mean and best context score. More negative context++ = stronger predicted repression. Defined on: arms in TargetScan's human default predictions (âš only 321)
ts_sites_per_gene ARM13%	TargetScan context++ scores for the arm's sites â€” number of sites, mean and best context score. More negative context++ = stronger predicted repression. Defined on: arms in TargetScan's human default predictions (âš only 321)

ISOC_ Â• 5

reference â€” context, provenance and rarely-quoted detail

isoc_canonical_frac ARM91%	isomiR COMPOSITION â€” what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm. Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_donated_frac ARM91%	isomiR COMPOSITION â€” what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm. Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_med_reads ARM91%	Median isomiR reads for this arm â€” median 1.00, max 528,927 .
isoc_n_dest_families ARM91%	isomiR COMPOSITION â€” what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm. Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss
isoc_orphan_frac ARM91%	isomiR COMPOSITION â€” what fraction of the arm's reads are canonical, orphan, or donated from a neighbouring arm. Defined on: arms in the isomiR seed-COMPOSITION table â€” where the reads go; â isoc_orphan_frac flags MH-153 mass loss

SDSZ_ 4

reference “ context, provenance and rarely-quoted detail

sdsz_N_7mer_plus ARM AGG EDGE 97%	MANE 3'UTR seed scan.
sdsz_N_8mer ARM AGG EDGE 97%	MANE 3'UTR seed scan.
sdsz_N_eff ARM AGG EDGE 97%	Effective seed-site count over the scanned universe (median 2,094, fill 96.7% “ the best-covered targetome axis).
sdsz_rarity ARM 97%	MANE 3'UTR seed scan.

FST_ 16

reference “ context, provenance and rarely-quoted detail

fst_d_share_HLY_TUM ARM 3%	As <code>fst_d_share_NAT_TUM</code> but GTEEx-healthy “ tumour, same common-support correction. n=68 (HLY is the sparsest leg at 17.0%). Post-fix mean ^'0.0000, 47.1% positive, p=0.93 .
fst_d_share_NAT_TUM ARM 6%	Change in the arm's share of its seed family's linear dose, matched NAT “ tumour, re-normalised over the members measured in BOTH states .
fst_dominance_switch ARM 7%	“ THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES “ gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).
fst_hly_family_measured ARM 100%	Do ALL members of this arm's family have a measured GTEEx level.
fst_hly_measured ARM 100%	“ THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES “ gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).
fst_is_dominant_HLY ARM 17%	“ THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES “ gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).
fst_is_dominant_TUM ARM 20%	“ THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES “ gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).
fst_n_common_HLY_TUM ARM 9%	Members measured in BOTH GTEEx-healthy and tumour “ the denominator behind <code>fst_d_share_HLY_TUM</code> . See <code>fst_n_common_NAT_TUM</code> .
fst_n_common_NAT_TUM ARM 20%	How many of the family's arms are measured in BOTH NAT and tumour “ the DENOMINATOR behind <code>fst_d_share_NAT_TUM</code> , shipped because the column was wrong for want of it.
fst_n_members ARM 100%	“ THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES “ gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).
fst_rank_HLY ARM 17%	“ THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES “ gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).
fst_rank_NAT ARM 74%	“ THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES “ gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).

fst_rank_TUM ARM20%	â THE ARM'S SHARE OF ITS OWN SEED FAMILY, ACROSS STATES â€” gene-free (<code>card_ladders.family_state_shift</code> , 2026-08-19).
fst_share_HLY ARM17%	Share of family dose in GTEx-healthy, from <code>hly_gtex_median</code> (fill 17.0% â€” the sparsest leg). Same singleton degeneracy; and cross-platform, so contrast only.
fst_share_NAT ARM74%	Share of family dose in matched NAT, from <code>hly_nat_median</code> (fill 74.0% â€” the best-covered leg, 3.7Å– TUM). Same singleton degeneracy: median 1.000, mask on <code>fst_n_members > 1</code> .
fst_share_TUM ARM20%	The arm's fraction of its seed family's total linear dose in tumour, from <code>arm_med_rpm</code> (fill 19.8%).

CHIM_ 5
reference â€” context, provenance and rarely-quoted detail

chim_manakov_n_genes ARMAGG EDGE60%	Chimeric-evidence ROLLUP for the arm (aggregated over its edges).
chim_manakov_weight ARMAGG EDGE60%	Chimeric-evidence ROLLUP for the arm (aggregated over its edges).
chim_n_sources ARMAGG EDGE63%	Chimeric-evidence ROLLUP for the arm (aggregated over its edges).
chim_tarbase_n_genes ARMAGG EDGE21%	Chimeric-evidence ROLLUP for the arm (aggregated over its edges).
chim_tarbase_weight ARMAGG EDGE21%	Chimeric-evidence ROLLUP for the arm (aggregated over its edges).

FIELD_ 5
reference â€” context, provenance and rarely-quoted detail

field_n ARM23%	Within-patient NAT field-effect fit â€” is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset. Defined on: arms with a within-patient NAT field-effect fit (571)
field_r_adj ARM23%	Within-patient NAT field-effect fit â€” is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset. Defined on: arms with a within-patient NAT field-effect fit (571)
field_r_own ARM23%	Within-patient NAT field-effect fit â€” is the regulator already established in the patient's adjacent normal before the tumour. Defined on the paired subset. Defined on: arms with a within-patient NAT field-effect fit (571)
field_r_perm ARM23%	The permutation null for <code>field_r_own</code> â€” median 0.000 , range [$\hat{a}^{0.11}$, +0.09].
field_retention ARM	â”â”â” NOT A RETENTION RATIO â€” a FIFTH sense of the word in this codebase. Measured: it equals <code>field_r_own â” field_r_perm</code> (exact within rounding on 100% of 571 arms, max residual 0.001), i.e.

SUB_ 6

reference context, provenance and rarely-quoted detail

sub_best_frac ARM30%	PAM50 subtype block aggregated to the ARM. See esub_ for the per-edge form. Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label
sub_best_mode ARM30%	PAM50 subtype block aggregated to the ARM. See esub_ for the per-edge form. Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label
sub_med_rho_best ARM30%	Median coupling in the arm's BEST subtype.
sub_med_rho_pool ARM30%	Median pooled-subtype coupling (median 0.0064) the unselected reference for sub_med_rho_best . Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label
sub_n_edges ARM30%	PAM50 subtype block aggregated to the ARM. See esub_ for the per-edge form. Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label
sub_n_q05 ARM30%	PAM50 subtype block aggregated to the ARM. See esub_ for the per-edge form. Defined on: arms with a per-PAM50 edge-heterogeneity fit (728) DISTRIBUTIONAL (MH-165), not a validated label

SEQ_ 4

reference context, provenance and rarely-quoted detail

seq_n_genes_any ARMAGG EDGE30%	Genes with ANY scanMiR K_D site
seq_n_genes_strong ARMAGG EDGE30%	Genes with a STRONG scanMiR K_D site median 3,632, spearman +0.822 with sdsz_N_eff and +0.797 with site_n_genes_canonical , but only +0.566 with ts_n_genes .
seq_promisc ARM30%	SEQUENCE targetome degree genes with a strong predicted duplex (scanMiR RBNS K_D), breast-expressed.
seq_promisc_any ARM30%	SEQUENCE targetome degree genes with a strong predicted duplex (scanMiR RBNS K_D), breast-expressed.

COV_ 22

reference context, provenance and rarely-quoted detail

cov_ago_dom ARM100%	Coverage flags was this row ever SCANNED by the named source.
cov_attr ARM100%	Coverage flags was this row ever SCANNED by the named source.
cov_beta ARM100%	Coverage flags was this row ever SCANNED by the named source.
cov_cnv ARM100%	Coverage flags was this row ever SCANNED by the named source.

cov_comp ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_cptac ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_curated ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_expr ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_famrole ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_field ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_focal ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_fst ARM100%	Was the family-share block computed for this arm at all “ i.e. did it have the abundance input.
cov_gctx ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_isocomp ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_isomir ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_meth ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_real ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_seed_scan ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_seq_scan ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_site_types ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_subtype ARM100%	Coverage flags “ was this row ever SCANNED by the named source.
cov_targetscan ARM100%	Coverage flags “ was this row ever SCANNED by the named source.

CUR_ 2

reference " context, provenance and rarely-quoted detail

cur_he_degree ARMAGG EDGE36%	How many HE genes this arm regulates in the curated design " the arm's curation degree (median 3, max 125). Defined on: arms with miRTarBase curation / ledger PMIDs " " FAME axes, not targetome (MH-208)
cur_he_degree_expr ARMAGG EDGE36%	As cur_he_degree but counting only targets above the expression floor.

ISO_ 5

reference " context, provenance and rarely-quoted detail

iso_n_samples ARM28%	isomiR profile for the arm " 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional). Defined on: arms with a 5'-isomiR seed-shift measurement (687) " " gate on iso_n_samples, the median is n-DEPENDENT
iso_seed_shift_frac ARM28%	isomiR profile for the arm " 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional). Defined on: arms with a 5'-isomiR seed-shift measurement (687) " " gate on iso_n_samples, the median is n-DEPENDENT
iso_seed_shift_sd ARM28%	isomiR profile for the arm " 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional). Defined on: arms with a 5'-isomiR seed-shift measurement (687) " " gate on iso_n_samples, the median is n-DEPENDENT
iso_top_shift_nt ARM28%	isomiR profile for the arm " 5' shift fraction and seed-shift occurrence. Drives the isomiR-corrected X_fam, which is BUILT but default-OFF (coupling wash, bidirectional). Defined on: arms with a 5'-isomiR seed-shift measurement (687) " " gate on iso_n_samples, the median is n-DEPENDENT
iso_total_rpm ARM	" A SUM ACROSS SAMPLES, not a per-sample RPM " the name misleads. Max 270,988,199, which is impossible for a per-million unit; divided by iso_n_samples the scale is sane (median 57, max 251,381).

FAMROLE_ 6

reference " context, provenance and rarely-quoted detail

famrole_abund_rank ARM91%	The arm's ROLE inside its seed family " member count, its share of family abundance, and its rank among members. Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_abund_share ARM91%	The arm's ROLE inside its seed family " member count, its share of family abundance, and its rank among members. Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1
famrole_class ARM100%	The arm's role in its seed family " sole on 1,639 of 2,450 (66.9%), dominant 282, co and the rest below.
famrole_is_dominant ARM91%	The arm's ROLE inside its seed family " member count, its share of family abundance, and its rank among members. Defined on: arms with a seed family " the arm's ROLE inside it; " DEGENERATE at famrole_n_members==1

famrole_n_members ARM 100%	The arm's ROLE inside its seed family â€™ member count, its share of family abundance, and its rank among members. Defined on: arms with a seed family â€™ the arm's ROLE inside it; â€™ DEGENERATE at famrole_n_members==1
famrole_var_rank ARM 88%	The arm's ROLE inside its seed family â€™ member count, its share of family abundance, and its rank among members. Defined on: arms with a seed family â€™ the arm's ROLE inside it; â€™ DEGENERATE at famrole_n_members==1

SURV_ 8

reference â€™ context, provenance and rarely-quoted detail

surv_dfi_hr_adj ARM 6%	Adjusted DFI hazard ratio per arm â€™ range [0.84, 1.16] , median 1.00 . The whole distribution sits within Â±16% of no effect. See <code>surv_dfi_q_adj</code> . Defined on: arms in the TCGA outcome panel (145) â€™ â€™ TCGA legs only; the GSE leg is bare-stem matched
surv_dfi_hr_base ARM 6%	TCGA outcome panel legs.
surv_dfi_p_adj ARM 6%	TCGA outcome panel legs.
surv_dfi_p_base ARM 6%	TCGA outcome panel legs.
surv_dfi_q_adj ARM 6%	BH q for the arm's DFI hazard ratio.
surv_os_hr_adj ARM 6%	TCGA outcome panel legs.
surv_os_p_adj ARM 6%	TCGA outcome panel legs.
surv_os_q_adj ARM 6%	TCGA outcome panel legs.

AID_ 4

reference â€™ context, provenance and rarely-quoted detail

aid_frac_resolvable ARM AGG ARM-IN-FAMILY 6%	Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates (<code>arm_resolvable</code> , <code>arm_sep_z</code> , <code>oof_drho</code> medians). Defined on: arms inside â€™1 MULTI-ARM (gene,family) cell â€™ the arm rung's 20.3%-of-edges footprint (142)
aid_med_arm_sep_z ARM AGG ARM-IN-FAMILY 6%	Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates (<code>arm_resolvable</code> , <code>arm_sep_z</code> , <code>oof_drho</code> medians). Defined on: arms inside â€™1 MULTI-ARM (gene,family) cell â€™ the arm rung's 20.3%-of-edges footprint (142)
aid_med_oof_drho ARM AGG ARM-IN-FAMILY 6%	Arm-IDENTIFIABILITY rollup for the arm: how often this arm is separable from its same-seed mates (<code>arm_resolvable</code> , <code>arm_sep_z</code> , <code>oof_drho</code> medians). Defined on: arms inside â€™1 MULTI-ARM (gene,family) cell â€™ the arm rung's 20.3%-of-edges footprint (142)

aid_n_cells	Multi-arm cells this arm participates in are fill 5.8% (the arm identifiability block is the sparsest on the card).
ARM	AGG ARM-IN-FAMILY 6%

BC_ 11
reference are context, provenance and rarely-quoted detail

bc_fam_context_homogeneous	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the seed FAMILY (genomic_context.family_context)
ARM 36%	
bc_fam_dose_coding_host_fra c	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the seed FAMILY (genomic_context.family_context)
ARM 36%	
bc_fam_dose_cotranscribed_f rac	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the seed FAMILY (genomic_context.family_context)
ARM 36%	
bc_fam_n_distinct_coding_ho sts	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the seed FAMILY (genomic_context.family_context)
ARM 36%	
bc_fam_n_members	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the seed FAMILY (genomic_context.family_context)
ARM 36%	
bc_meth_dbeta	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the pri-miRNA HAIRPIN are 5p and 3p share one promoter and one are (354 arms)
ARM 14%	
bc_meth_direction	Tumour-vs-normal methylation direction at this arm's locus.
ARM 14%	
bc_meth_n_cgi	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the pri-miRNA HAIRPIN are 5p and 3p share one promoter and one are (354 arms)
ARM 14%	
bc_meth_n_probes	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the pri-miRNA HAIRPIN are 5p and 3p share one promoter and one are (354 arms)
ARM 14%	
bc_meth_normal_beta	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the pri-miRNA HAIRPIN are 5p and 3p share one promoter and one are (354 arms)
ARM 14%	
bc_meth_tumour_beta	Breast-cancer locus annotation for the arm are CpG probes, CGI overlap and related epigenetic context at the precursor locus. Defined on: are BROADCAST from the pri-miRNA HAIRPIN are 5p and 3p share one promoter and one are (354 arms)
ARM 14%	

WSHIFT_ 6

reference “ context, provenance and rarely-quoted detail

wshift_mean_dGlobalRank	Within-patient paired quantities on the ~103 matched tumour/NAT participants.
ARM 52%	
wshift_mean_own_shift	Within-patient paired quantities on the ~103 matched tumour/NAT participants.
ARM 52%	
wshift_n_pairs	Within-patient paired quantities on the ~103 matched tumour/NAT participants.
ARM 91%	
wshift_nat_patient_sd	Within-patient paired quantities on the ~103 matched tumour/NAT participants.
ARM 74%	
wshift_own_specific_frac	Within-patient paired quantities on the ~103 matched tumour/NAT participants.
ARM 45%	
wshift_sd_dGlobalRank	Within-patient paired quantities on the ~103 matched tumour/NAT participants.
ARM 45%	

TIER_ 5

reference “ context, provenance and rarely-quoted detail

tier_class	Expression tier “ silent on 1,518 of 2,236 arms (67.9%), conditional 476, the rest robust.
ARM 91%	
tier_expressed	Expression tier of the arm against the detection floor “ class, fraction of samples expressed, p90 and max log2 RPM. Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
ARM 91%	
tier_frac_expressed	Expression tier of the arm against the detection floor “ class, fraction of samples expressed, p90 and max log2 RPM. Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
ARM 91%	
tier_max	Expression tier of the arm against the detection floor “ class, fraction of samples expressed, p90 and max log2 RPM. Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
ARM 91%	
tier_p90	Expression tier of the arm against the detection floor “ class, fraction of samples expressed, p90 and max log2 RPM. Defined on: arms detected in the TCGA miRNA matrix (2,236 of 2,450)
ARM 91%	

DISC_ 2

reference “ context, provenance and rarely-quoted detail

disc_gold_families	Which seed families this row's discovery-queue edges belong to, semicolon-separated. Usually ONE at the arm rung (an arm has one seed).
ARM 1%	
disc_n_gold_edges	How many of the 157 discovery-queue edges name this arm. Concentrated: miR-106b-5p 39 Â· miR-20a-5p 25 Â· miR-93-5p 25 Â· miR-19a-3p 19 Â· miR-18a-5p 9.
ARM 100%	

CNV_ 9

reference “ context, provenance and rarely-quoted detail

cnv_mean_cn ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>
cnv_median_cn ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>
cnv_pct_altered ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>
cnv_pct_amp ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>
cnv_pct_deep_del ARM 35%	Percent of samples with a deep deletion at this arm's locus.
cnv_pct_gain ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>
cnv_pct_loh ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>
cnv_pct_loss ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>
cnv_sd_cn ARM 35%	Copy number at the arm's own locus “ mean CN and amplification/deletion summaries. <i>Defined on: arms with a mapped CN locus in the cohort stratum (856) “ “ multi-locus arms are DILUTED here</i>

GCTX_ 16

reference “ context, provenance and rarely-quoted detail

gctx_clustered ARM 29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms “ 714)</i>
gctx_context_mixed ARM 29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms “ 714)</i>
gctx_coord_source ARM 29%	Provenance of the genomic-context coordinates.
gctx_host ARM 27%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms “ 714)</i>
gctx_host_region ARM 27%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (genomic_context.classify_he_arms “ 714)</i>

gctx_host_type ARM27%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_in_exon ARM29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_mir_class ARM29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_mirgenedb ARM29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_n_loci ARM29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_promoter_class ARM29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_promoter_coord_class ARM29%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_promoter_gene ARM24%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_promoter_gene_type ARM24%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_promoter_host_tss_kb ARM24%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>
gctx_strand_rel ARM27%	Genomic context of the arm's precursor locus (<code>genomic_context.classify_he_arms</code>): host gene, sense/antisense orientation, intron-vs-exon, promoter class. Strand-aware. <i>Defined on: arms with a classified precursor locus (<code>genomic_context.classify_he_arms</code> 714)</i>

SURROGATE_ 2

reference context, provenance and rarely-quoted detail

surrogate_corr ARM0%	Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for. <i>Defined on: arms with a same-seed SURROGATE (MH-166) 3.6% of edges</i>
surrogate_instrument ARM0%	Surrogate healthy instrument used where a direct healthy leg is unavailable, and its correlation with the quantity it stands in for. <i>Defined on: arms with a same-seed SURROGATE (MH-166) 3.6% of edges</i>

CNVC_ 5

reference “ context, provenance and rarely-quoted detail

cnvc_n ARM 31%	CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.
cnvc_partial_p ARM 31%	CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.
cnvc_partial_q ARM 31%	BH q for the arm's CN-coupling partial correlation.
cnvc_partial_rho ARM 31%	CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.
cnvc_spearman_rho ARM 31%	CN-to-expression concordance fit for the arm (756 arms): does locus CN actually move this arm's abundance.

FOC_ 7

reference “ context, provenance and rarely-quoted detail

foc_focal_locus ARM 30%	Paralog-decontaminated FOCAL-locus concordance.
foc_multilocus ARM 30%	Paralog-decontaminated FOCAL-locus concordance.
foc_n ARM 30%	Paralog-decontaminated FOCAL-locus concordance.
foc_n_loci ARM 30%	Paralog-decontaminated FOCAL-locus concordance.
foc_partial_p ARM 30%	Paralog-decontaminated FOCAL-locus concordance.
foc_partial_rho ARM 30%	Paralog-decontaminated FOCAL-locus concordance.
foc_spearman_rho ARM 30%	Paralog-decontaminated FOCAL-locus concordance.

MODEL_ 1

reference “ context, provenance and rarely-quoted detail

model_n_genes ARM AGG EDGE 30%	Which model variant produced this row. Defined on: arms carrying >=1 curated HE edge in the model design (741)
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COMV_ 2

reference context, provenance and rarely-quoted detail

comv_module ARM15%	Co-expression module membership and top co-expressed partner. Pure co-expression the cross-state class columns are deliberately excluded from this block. Defined on: arms in a co-expression module (371) pure co-expression; the state-class columns are excluded
comv_top_partner ARM15%	Co-expression module membership and top co-expressed partner. Pure co-expression the cross-state class columns are deliberately excluded from this block. Defined on: arms in a co-expression module (371) pure co-expression; the state-class columns are excluded

CLOAD_ 3

reference context, provenance and rarely-quoted detail

cload_dose_rpm ARM60%	Compartment LOADING of the arm.
cload_lock ARM8%	Compartment LOADING of the arm.
cload_top_compartment ARM60%	Compartment LOADING of the arm.

HX_ 1

reference context, provenance and rarely-quoted detail

hx_hly_baseline_imputed ARM28%	IMPUTED healthy baseline (miTED rank-transfer or seed-mate substitution) LABELLED as imputed, never presented as a measurement.
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seed_family_card.tsv

seed_family

One row per the seed family itself, gene-free. 83.7% of families are SINGLETONS for those every 'family-level' statistic is trivially that one arm's value. Mask on fam_single_member first.

33 columns 1,959 rows 2 blocks median coverage 92%

Contents 2 blocks, most important first

1. seed_1 CORE
2. fam_32 REFERENCE

SEED_1

core the columns findings rest on

seed_family	KEY the seed family itself, one row per family.
KEY	100%

FAM_32

reference context, provenance and rarely-quoted detail

fam_ctx_composition	The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.
FAMILY	29%
fam_ctx_context_homogeneous	The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.
FAMILY	29%
fam_ctx_dose_coding_host_frac	The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.
FAMILY	29%
fam_ctx_dose_cotranscribed_frac	The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.
FAMILY	29%
fam_ctx_n_distinct_coding_hosts	The family's GENOMIC context, pooled over its members: host-coding fraction, co-transcribed fraction, distinct coding hosts, and whether the members share a context. Fill 29.1%.
FAMILY	29%
fam_cur_degree_max	Seed-family structure the arm sits in membership, affinity and dominance within it.
FAMILY	36%
fam_dominant_arm	Seed-family structure the arm sits in membership, affinity and dominance within it.
FAMILY	92%
fam_dominant_share	The top member's share of the family's total dose.
FAMILY	92%

fam_dose_hhi FAMILY 15%	Concentration of DOSE across the family's members. â€¦ FLOOR-CORRECTED â€” verified 2026-08-19 : its minimum sits far below the raw 1/k floor at every member count (0.0000 at k=2 where the raw floor is 0.500), and spearman(n_members, hhi) = +0.1896 rather than the strongly negative value a raw index would give.
fam_dose_max_log2 FAMILY 92%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_dose_med_log2 FAMILY 92%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_dose_total_log2 FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_fame_npmid_sum FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_n_expressed FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_n_guide FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_n_members FAMILY 100%	Arms in the seed family (median 1, max 22).
fam_n_members_context FAMILY 29%	Arms in the seed family (median 1, max 22).
fam_n_model_edges FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_n_passenger FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_n_shift_gt20_gated FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_real_n_c10_sum FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_real_n_scored_sum FAMILY 100%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_redund_m_eff FAMILY 4%	Within-family redundancy â€” effective member count, over-count fraction, and the median within-family correlation.
fam_redund_median_within_rho FAMILY 4%	Within-family redundancy â€” effective member count, over-count fraction, and the median within-family correlation.
fam_redund_overcount_frac FAMILY 4%	Within-family redundancy â€” effective member count, over-count fraction, and the median within-family correlation.
fam_seed_shift_max_gated FAMILY 13%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.

fam_single_member FAMILY 100%	â€” THE DOMINANT FACT ABOUT THIS ENTIRE CARD: 1,639 of 1,959 families (83.7%) are SINGLETONS. For those rows every 'family-level' statistic is TRIVIALY that one arm's value, and fam_dominant_share is forced to exactly 1.0.
fam_targetome_seed_med FAMILY 97%	Family-level targetome size.
fam_targetome_seq_med FAMILY 30%	Family-level targetome size.
fam_var_hhi FAMILY 14%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_var_max FAMILY 89%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.
fam_var_med FAMILY 89%	Seed-family structure the arm sits in â€” membership, affinity and dominance within it.